

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Bismillah-ir-Rahman-ir-Raheem

Niyat, Mehnat aur Allah par Bharosa

Ilm woh roshni hai jo neeyat ko sache rakhe to har mushkil aasan kar deti hai.

Mehnat kabhi rayegan nahi jati, Allah har koshish ka sila zaroor deta hai.

Consistency wahi rakhta hai jo apne maqsad par yaqeen rakhta hai.

Har raat ki padhai, kal ki kamyabi ki bunyaad hai.

Allah par tawakkul rakho, mehnat tumhara kaam hai, natija Uska.

Jo aaj thoda thoda karta hai, woh kal bohot kuch hasil karta hai.

"Wa man yatawakkal 'alallahi fahuwa hasbuh" — jo Allah par bharosa kare, Allah kaafi hai.

MONEY AND BANKING · COMPLETE STUDY NOTES

MGT411

Money and Banking

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LESSON 19 – LESSON 45 · COMPREHENSIVE LECTURE NOTES · PREMIUM EDITION

Table of Contents

A complete journey through stocks, banking, and monetary policy

Lesson 19 Valuing Stocks	3
Lesson 20 Risk and Value of Stocks	4
Lesson 21 Role of Financial Intermediaries	5
Lesson 22 Role of Financial Intermediaries (Continued)	6

Lesson 23 Banking	7
Lesson 24 Balance Sheet of Commercial Banks	8
Lesson 25 Bank Risk	9
Lesson 26 Interest Rate Risk (Cont.) and Globalization of Banking	10
Lesson 27 Non-Depository Institutions	11
Lesson 28 Non-Depository Institutions (Continued) and Banking Crisis	12
Lesson 29 The Government Safety Net	13
Lesson 30 The Government's Bank	14
Lesson 31 Low, Stable Inflation and Central Bank Objectives	15
Lesson 32 Creating a Successful Central Bank	16
Lesson 33 The Monetary Base	17
Lesson 34 Deposit Creation in a Single Bank	18
Lesson 35 Money Multiplier and the Monetary Policy Toolbox	19
Lesson 36 Target Federal Funds Rate and Open Market Operations	20
Lesson 37 Why Do We Care About Monetary Aggregates?	21
Lesson 38 The Facts About Velocity	22
Lesson 39 The Portfolio Demand for Money	23
Lesson 40 Money Growth, Inflation, and Aggregate Demand	24
Lesson 41 Deriving the Monetary Policy Reaction Curve	25
Lesson 42 The Aggregate Demand Curve	26
Lesson 43 The Aggregate Supply Curve	27
Lesson 44 Equilibrium and the Determination of Output and Inflation	28
Lesson 45 Shifts in Potential Output and Real Business Cycle Theory	29

LESSON 19

Valuing Stocks

Deep Explanation

Yeh bilkul waisi baat hai jaise tum ek dukaan mein partnership karte ho — agar dukaan acha chalti hai tou tumhara profit zyada hota hai, magar agar dukaan mein nuqsan ho jaye tou pehle bank ka loan wapas dena parta hai, aur jo bachta hai wahi tumhara hota hai. Stocks bhi waisi hi hoti hain, jahan shareholders ko sabse last mein paisay milte hain.

Key Terms

1. Chartist: Yeh log past price patterns dekh kar future price predict karne ki koshish karte hain.

- 2. Behavioralist:** Yeh log investor psychology aur behavior ke through stock ki value estimate karte hain.
- 3. Fundamental Value:** Yeh value firm ke financial statements aur future profitability ke detailed analysis se nikalti hai.
- 4. Dividend:** Yeh firm ke profits ka woh hissa hai jo shareholders ko diya jata hai.
- 5. Dividend Discount Model:** Yeh formula stock ki price ko uske future dividends aur growth rate ke through calculate karta hai.
- 6. Leverage:** Jab company apna business debt (bonds) se finance karti hai tou yeh risk create karta hai jo equity holders pe ata hai.
- 7. Residual Claimant:** Stockholders ko sabse last mein payment milti hai, sab dusre logon (jaise bondholders) ke baad.

$$P_{\text{today}} = D_{\text{next year}} / (1+i) + P_{\text{next year}} / (1+i)$$

$$P_{\text{today}} = D_{\text{today}} / (i - g)$$

Key Points

- 1. Stock Price Formula:** Aaj ki stock price uske future dividends aur agle saal ki selling price ka present value hoti hai.
- 2. Constant Growth Assumption:** Agar dividends har saal ek fixed rate (g) se grow karte hain, tou price ek simple formula se nikal sakti hai.
- 3. Consol Comparison:** Agar firm hamesha dividends deti rahe tou stock ko ek consol bond jaisa treat kiya ja sakta hai, jahan Price = Dividend / (i - g).
- 4. High Price Conditions:** Stock price zyada hoti hai jab dividends high hon, growth rate fast ho, ya interest rate kam ho.
- 5. Risk Source:** Stocks risky hoti hain kyunke shareholders ko firm ke sab dusre payments (jaise bonds) ke baad paisay milte hain, isliye unka return uncertain hota hai.

Equity %	Debt %	Bond Payment	Equity Return %	St. Dev. of Return
100%	0	—	8–16%	4%
50%	50%	\$50	6–22%	8%
30%	70%	\$70	3.3–30%	13.3%
20%	80%	\$80	0–40%	20%

Example

Socho ek dost apna chota sa mobile repair shop khol raha hai. Usne kuch paisay khud lagaye (equity) aur kuch bank se loan liya (debt) taake ek machine khareed sake. Agar business acha chala tou loan ka interest pay karne ke baad jo bacha woh sab uska profit hai, magar agar business slow raha tou pehle bank ko paisay dene parenge, aur dost ko bohot kam ya kabhi kabhi kuch bhi na mile. Yahi waja hai ke equity (stock jaisi) investment debt se zyada risky hoti hai.

Conceptual Questions

Q: Why are stocks considered riskier than bonds, even though both are forms of firm financing?

Answer: Bonds wale logon ko fixed payment milti hai aur unhe pehle paisay milte hain, magar stockholders ko sabse aakhir mein jo bacha hota hai wahi milta hai.

Jaise upar wali table mein dekho, jab equity kam hoti hai (sirf 20%) tou expected return zyada hota hai (20%) magar uski volatility (St. Dev.) bhi bohot zyada hoti hai (20%), kyunke saara risk equity holders pe ajata hai.

Q: What is the difference between how Chartists/Behavioralists value stocks versus how fundamental analysts value stocks?

Answer: Chartists aur Behavioralists price ke patterns ya investor psychology dekh kar value nikalte hain, jabke fundamental analysts firm ke actual financial statements aur future profitability dekh kar real value calculate karte hain.

LESSON QUIZ

1. What do stock payments to investors consist of? **(Dividends)**
2. What does the Dividend Discount Model state stock price equals? **(Dividend divided by interest rate minus growth rate)**
3. What happens to stock price when dividend growth rate increases? **(It increases)**
4. What creates risk for equity holders when a firm uses debt financing? **(Leverage)**
5. Why are stockholders called residual claimants? **(Because they are paid last, after all other obligations are met)**

LESSON 20

Risk and Value of Stocks

Deep Explanation

Yeh bilkul waisi baat hai jaise tum ek dukaan kholne se pehle sirf rent aur saman ki cost nahi dekhte, balke yeh bhi sochte ho ke risk kitna hai aur uska reward kitna milna chahiye. Stocks mein bhi investors sirf dividends nahi dekhte, woh risk ka extra muawza (compensation) bhi maangte hain.

Key Terms

1. **Risk Premium:** Yeh extra return hai jo investors stock ke risk lene ke badle mein maangte hain.
2. **Risk-free Return:** Yeh woh return hai jo bilkul safe investment (jaise treasury security) se milta hai.
3. **Required Stock Return:** Yeh risk-free return aur risk premium ko milakar banta hai.
4. **Efficient Markets Theory:** Yeh theory kehti hai ke stock prices already sari available information ko reflect karti hain.
5. **Bubble:** Yeh stock prices aur unki actual fundamental value ke darmiyan ek barhta hua farq hai.
6. **Financial Intermediary:** Yeh institution hai jo savers ka paisa pool karke borrowers ko lend karti hai, jaise banks.
7. **Direct Finance:** Is mein borrowers seedha lenders ko securities bechte hain, bina kisi beech wale ke.
8. **Indirect Finance:** Is mein ek third party (jaise bank) lenders aur borrowers ke darmiyan kaam karti hai.

$$\text{Required Stock Return (i)} = \text{Risk-free Return (r}_f\text{)} + \text{Risk Premium (r}_p\text{)}$$

$$P_{\text{today}} = D_{\text{today}} / (r_f + r_p - g)$$

Key Points

1. **Adjusted Dividend Discount Model:** Stock price ka formula ab risk-free rate aur risk premium ko bhi include karta hai.
2. **High Price Conditions:** Stock price high hoti hai jab dividends high hon, growth fast ho, aur risk-free rate ya risk premium kam ho.
3. **Long Run vs Short Run:** Stocks short term mein risky hoti hain, magar long term mein unka risk bonds se bhi kam ho jata hai.
4. **Stock Market's Role:** Stock prices firms ke market value batati hain, jo resources ki allocation decide karti hain.
5. **Information Problem:** Online stores mortgages ya loans nahi bechte kyunke unmein information aur transaction costs bohot zyada hoti hain, isliye financial intermediaries zaroori hain.

Example

Socho tum apne dost ko paisay lend karna chahte ho taake woh apna chota business shuru kare. Tumhe pata nahi ke woh paisay wapas karega ya nahi, isliye tum extra return maangte ho jaise "agar wapas kiya tou zyada dena". Yahi risk premium ki tarah hai jo stock investors company se maangte hain. Lekin agar 100 log mil kar yeh kaam karna chahein tou records rakhna, contracts banana, aur creditworthiness check karna mushkil ho jata hai — isliye bank jaisi intermediary zaroori hoti hai.

Conceptual Questions

Q: Why is the risk-free rate and risk premium added to the dividend discount model instead of just using the original formula?

Answer: Original formula sirf dividend growth aur interest rate ko dekhti thi, magar stock ki future selling price uncertain hoti hai, isliye investors ko risk ka extra muawza chahiye hota hai jo risk premium ke through add kiya jata hai.

Jaise 2003 mein actual stock prices calculated value se zyada thi, jiski waja yeh nikli ke investors ne kam risk premium demand ki thi, na ke dividend ya growth rate change hui thi.

Q: Why do financial intermediaries play a bigger role in the economy than direct finance through stock and bond markets?

Answer: Indirect finance information aur transaction costs ko kam karta hai, jo direct finance mein bohot zyada hoti hain.

Jaise online store easily \$300,000 ki car bech sakta hai magar mortgage nahi de sakta, kyunke usay borrower ki creditworthiness check karni paregi, legal contracts banane parenge, aur payments collect karni parengi — yeh sab kaam bank zyada acha kar sakta hai.

LESSON QUIZ

1. What two components make up the Required Stock Return? (**Risk-free return and risk premium**)
2. According to Efficient Markets Theory, what do current stock prices reflect? (**All available information**)
3. Are stocks more or less risky when held for the long term compared to bonds? (**Less risky**)
4. What is a persistent gap between actual stock prices and fundamental values called? (**A bubble**)
5. Name one function performed by financial intermediaries. (**Pooling savings, providing liquidity, diversifying risk, or reducing information costs**)

LESSON 21

Role of Financial Intermediaries

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apna pocket money ek mutual friend ke pass jama karte ho jo sabki savings collect karke unhe trustworthy logon ko lend karta hai, aur saath mein hisaab kitab bhi rakhta hai. Financial intermediaries bhi yeh hi kaam karte hain, sirf bohot bare scale par.

Key Points

- 1. Indirect Finance Importance:** Financial intermediaries ke through hone wala indirect finance, direct finance (stock aur bond markets) se zyada important hota hai.
- 2. Information Reduction:** Intermediaries lending aur borrowing ki transaction aur information costs ko kam karte hain.

- 3. Pooling Savings:** Intermediaries bohut se chote savers ka paisa pool karte hain aur unhe apni soundness ka yaqeen dilate hain.
- 4. Safekeeping Origin:** Goldsmiths se shuru hua tha jab log apna sona unke pass rakhte the aur receipt lete the, jo aage paisay ki tarah use hone lagi.
- 5. Payments Access:** Banks ATMs, checkbooks, aur statements ke through logon ko payments system tak access dete hain.
- 6. Comparative Advantage:** Log apni best activity par focus karte hain jab unhe pata hota hai ke transaction costs kam hain, jo specialization aur trading ko barhata hai.
- 7. Economies of Scale:** Jab production zyada hoti hai tou average cost kam ho jati hai, aur yeh information par bhi lagu hota hai.

Types

- 1. Pooling Savings:** Choti savings ko collect karke barre investments ke liye use karna.
- 2. Safekeeping and Payments Access:** Paisay ko mehfooz rakhna aur payment services dena.
- 3. Liquidity:** Asaani se paisay ko cash mein convert karna.
- 4. Risk Diversification:** Risk ko alag alag jagah spread karna.
- 5. Information Services:** Borrower ki creditworthiness check karna aur information costs kam karna.

Example

Socho tumhare mohalle mein ek dukaandar hai jo logon se thora thora paisa jama karta hai aur phir us paisay se zaroratmand logon ko loan deta hai, jaise kisi ko apni dukaan banane ke liye. Woh ye bhi check karta hai ke borrower paisay wapas kar sakta hai ya nahi, aur sab ka hisaab bhi rakhta hai. Yahi ek financial intermediary ka kaam hai jo bank bare scale par karta hai.

Conceptual Questions

Q: Why is indirect finance through financial intermediaries more dominant globally than direct finance through stock and bond markets?

Answer: Indirect finance mein intermediaries lending aur borrowing ki information aur transaction costs ko kam kar dete hain, jo direct finance mein har lender ko khud uthani parti hain.

Jaise agar tum khud kisi ko paisay lend karo, tumhe uski creditworthiness khud check karni paregi, magar bank yeh kaam asaani se kar leta hai kyunke unke pass already systems aur expertise hoti hai.

Q: How does the principle of comparative advantage relate to the role of financial intermediaries?

Answer: Jab transaction costs kam hoti hain (jo intermediaries ki waja se hoti hain), tou log apni best activity par focus kar sakte hain bajaye is ke ke woh khud apne financial maamlat sambhalein.

Jaise ek doctor apna time patients dekhne mein lagata hai aur apni savings ko bank mein manage karwata hai, bajaye is ke ke woh khud loans dena ya invest karna seekhe.

LESSON QUIZ

1. Name the five functions of financial intermediaries. **(Pooling savings, safekeeping/payments access, liquidity, risk diversification, information services)**
2. Who were the original bankers? **(Goldsmiths)**
3. What helped people realize trading was easier than carrying gold? **(Receipts)**
4. What economic principle leads to specialization based on lower opportunity cost? **(Comparative advantage)**
5. What happens to average cost as production quantity increases, according to economies of scale? **(It falls/decreases)**

Role of Financial Intermediaries (Continued)

Deep Explanation

Yeh bilkul waisi baat hai jaise tum koi second-hand mobile khareedne se pehle uska check karwate ho ke kahin woh kharab tou nahi, kyunke seller ko us mobile ke baare mein tumse zyada pata hota hai. Financial markets mein bhi yeh hi problem hoti hai jab ek banda dusre se zyada information rakhta hai.

Key Terms

- 1. Liquidity:** Yeh batata hai ke kisi asset ko paise mein convert karna kitna asaan aur kam cost wala hai.
- 2. Diversification:** Apna paisa ek jagah lagane ke bajaye alag alag jagah lagana taake risk kam ho.
- 3. Information Asymmetry:** Jab ek party (jaise borrower) ke pass dusri party (jaise lender) se zyada information ho.
- 4. Adverse Selection:** Yeh problem transaction se pehle hoti hai jab acha ya bura credit risk pehchanna mushkil ho.
- 5. Moral Hazard:** Yeh problem transaction ke baad hoti hai jab pata nahi chalta ke borrower ne paisay sahi use kiye ya nahi.
- 6. Collateral:** Yeh woh cheez hai jo borrower lender ko pledge karta hai agar woh default kar jaye.
- 7. Net Worth:** Yeh firm ki value minus uski liabilities hoti hai, jo owner ka apna stake batati hai.
- 8. Principal-Agent Problem:** Yeh problem hoti hai jab company ke managers (agents) owners (principals) ke best interest mein kaam nahi karte.

Key Points

- 1. Liquidity Function:** Financial intermediaries asset ko kam cost mein paisay mein convert karne ki suhulat dete hain, jaise ATMs.
- 2. Risk Diversification:** Bank lakhon depositors ka paisa collect karke hazaron loans mein lagata hai, isliye har depositor ka risk bohot kam ho jata hai.
- 3. Lemons Problem:** Used car market mein jab buyers acha ya bura car nahi pehchan sakte, tou acchi cars market se gayab ho jati hain aur sirf kharab cars reh jati hain.
- 4. Solving Adverse Selection:** Information disclosure, collateral, aur net worth ke through good aur bad credit risks ko alag kiya ja sakta hai.
- 5. Solving Moral Hazard:** Reporting requirements aur restrictive covenants borrowers ko sahi tarah paisay use karne par majboor karte hain.

Questions Asked

- 1. What is liquidity and how do financial intermediaries provide it?** Yeh batata hai ke asset ko paisay mein badalna kitna asaan hai, aur intermediaries yeh kaam kam cost mein karte hain.
- 2. How does diversification reduce risk?** Paisay ko alag alag investments mein spread karne se ek jagah ke nuqsan ka asar kam ho jata hai.
- 3. What problems does information asymmetry create?** Yeh adverse selection aur moral hazard jaisi problems create karti hai.

Example

Socho tum kisi purani gari kharidne jaa rahe ho, magar tumhe nahi pata ke gari acchi hai ya kharab, isliye tum sirf average price dene ko tayyar ho. Iski waja se acchi gari ka malik woh price par bechna nahi chahega, aur sirf kharab gariyan market mein reh jayengi — yahi adverse selection hai. Isi tarah agar koi tumse loan le kar paisay kisi galat kaam mein lagaye, jis ka tumhe pata na chale, tou yeh moral hazard hai.

Conceptual Questions

Q: What is the key difference between adverse selection and moral hazard in financial transactions?

Answer: Adverse selection transaction se pehle hoti hai jab lender good aur bad credit risk mein farq nahi kar sakta, jabke moral hazard transaction ke baad hoti hai jab lender ko pata nahi chalta ke borrower paisay sahi use kar raha hai ya nahi.

Jaise loan dene se pehle borrower ki creditworthiness check karna adverse selection ka solution hai, magar loan dene ke baad uski monitoring karna moral hazard ka solution hai.

Q: Why does collateral help reduce the adverse selection problem in lending?

Answer: Collateral lender ko ek guarantee deta hai ke agar borrower default kar jaye tou unhe kuch compensation mil jayegi, isliye yeh ek tarah se borrower ki seriousness ko bhi zahir karta hai.

Jaise ghar ke loan mein agar koi default kare tou bank uska ghar le sakta hai, isliye banks home loans dene mein zyada confident hote hain bajaye unsecured loans ke.

LESSON QUIZ

1. What term describes when one party in a transaction has more information than the other? (**Information asymmetry**)
2. What problem occurs before a transaction takes place due to information asymmetry? (**Adverse selection**)
3. What problem occurs after a loan is given, when lenders cannot monitor borrower behavior? (**Moral hazard**)
4. What is the classic example used to explain adverse selection in markets? (**The market for lemons / used cars**)
5. Name one solution to the adverse selection problem. (**Collateral, net worth, or disclosure of information**)

LESSON 23

Banking

Deep Explanation

Yeh bilkul waisi baat hai jaise tumhare ghar mein kuch saman aaye (jo tumhara hai) aur kuch udhaar par liya hua hai, aur in dono ka farq batata hai ke tumhara actual net worth kya hai. Bank ka balance sheet bhi yeh hi kaam karta hai, jahan assets aur liabilities ka farq bank ka capital batata hai.

Key Terms

1. **Commercial Bank:** Yeh bank deposits leta hai aur unse consumer, commercial, aur real estate loans deta hai.
2. **Savings Institution:** Yeh intermediary households ko mortgage aur saving deposit services provide karta hai.
3. **Credit Union:** Yeh nonprofit institution hai jo common bond rakhne wale logon ka hota hai aur choti consumer loans deta hai.
4. **Bank Capital (Net Worth):** Yeh bank ke assets aur liabilities ka farq hota hai.
5. **Reserves:** Yeh cash hai jo bank apne vault ya central bank mein rakhta hai customer withdrawals ke liye.
6. **Secondary Reserves:** Securities ko yeh naam diya jata hai kyunke yeh highly liquid hoti hain aur jaldi cash mein convert ho sakti hain.
7. **Loans:** Yeh modern commercial banks ka sabse bara asset hai, jisme business, real estate, aur consumer loans shamil hain.

$$\text{Total Bank Assets} = \text{Total Bank Liabilities} + \text{Bank Capital}$$

Key Points

1. **Balance Sheet Identity:** Total Bank Assets hamesha Total Bank Liabilities aur Bank Capital ke barabar hote hain.
2. **Three Depository Institutions:** Commercial banks, savings institutions, aur credit unions teen basic types hain.
3. **Profit Sources:** Bank ka profit service fees aur interest earned minus interest paid se ata hai.
4. **Cash Opportunity Cost:** Cash par koi interest nahi milta, isliye banks apna cash holding kam rakhte hain.
5. **Loan Portfolio Difference:** Commercial banks business ko loans dete hain, savings institutions mortgages dete hain, aur credit unions consumer loans dete hain.

Superstructure

1. **Assets:** Yeh bank ke funds ka use hain, jaise cash, securities, aur loans.
2. **Liabilities:** Yeh bank ke funds ka source hain, jaise deposits aur borrowings.
3. **Bank Capital:** Yeh assets minus liabilities hota hai, jo bank ka apna stake hota hai.

Example

Socho tumhare pass ek choti dukaan hai. Tumne kuch saman apne paisay se khareeda (yeh tumhara capital hai) aur kuch saman udhaar par liya (yeh liability hai). Agar tum apni dukaan ke total saman (assets) se udhaar (liabilities) minus karo, jo bachega woh tumhara apna asal stake hai. Bank bhi waisi hi dukaan ki tarah hai, jahan deposits liability hain aur loans assets hain.

Conceptual Questions

Q: What is the key difference between how commercial banks, savings institutions, and credit unions use their loan portfolios?

Answer: Commercial banks zyada tar business ko loans dete hain, savings institutions individuals ko mortgages dete hain, aur credit unions choti consumer loans specialize karte hain.

Jaise agar tumhe ghar ke liye mortgage chahiye tou savings institution zyada appropriate hoga, magar agar tumhe apni dukaan ke liye loan chahiye tou commercial bank zyada acha rahega.

Q: Why do banks try to minimize the amount of cash they hold as reserves?

Answer: Cash par koi interest nahi milta, isliye jitna zyada cash bank apne pass rakhega, utna zyada uska opportunity cost hoga kyunke woh paisa loans ya securities mein laga kar profit kama sakta tha.

Jaise agar bank apna sara paisa vault mein rakh de tou woh customer withdrawals ke liye safe rahega, magar usay koi return nahi milega, isliye banks sirf zaroori amount hi reserve mein rakhte hain.

LESSON QUIZ

1. Name the three basic types of depository institutions. **(Commercial banks, savings institutions, credit unions)**
2. What is the Balance Sheet Identity for banks? **(Total Bank Assets = Total Bank Liabilities + Bank Capital)**
3. What is the difference between a bank's assets and liabilities called? **(Bank capital or net worth)**
4. Why are securities sometimes called secondary reserves? **(Because they are highly liquid and can be sold quickly for cash)**
5. What is the primary asset of modern commercial banks? **(Loans)**

Balance Sheet of Commercial Banks

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apne ghar ke kharchay chalane ke liye kabhi apni jama poonji use karte ho aur kabhi kisi dost se udhaar lete ho. Bank bhi apna kaam chalane ke liye deposits aur borrowings jaise sources use karta hai, aur in sab se woh apna profit nikalne ki koshish karta hai.

Key Terms

- 1. Checkable Deposits:** Yeh checking accounts hain jin par bank kam interest deta hai.
- 2. Nontransactions Deposits:** Yeh savings aur time deposits hain jo bank ki liabilities ka bara hissa hoti hain.
- 3. Certificate of Deposit (CD):** Yeh waisi hi hota hai jaise tum bank se bond khareed rahe ho.
- 4. Repurchase Agreement (Repo):** Yeh ek short-term loan hai jisme security cash ke against exchange hoti hai aur baad mein wapas li jati hai.
- 5. Bank Capital:** Yeh bank ke owners ka stake hota hai, jo sudden losses ke against cushion ka kaam karta hai.
- 6. Return on Assets (ROA):** Yeh batata hai ke bank apne assets ko kitni efficiency se use kar raha hai.
- 7. Return on Equity (ROE):** Yeh bank ke owners ko milne wala return measure karta hai.
- 8. Net Interest Margin:** Yeh bank ka interest rate spread hota hai, jo total assets ke percentage mein hota hai.
- 9. Letter of Credit:** Yeh bank ki guarantee hoti hai ke customer apna promised payment kar dega.

$$\text{ROA} = \text{Net Profit After Taxes} \div \text{Total Bank Assets}$$

$$\text{ROE} = \text{Net Profit After Taxes} \div \text{Bank Capital}$$

$$\text{ROE} = \text{ROA} \times (\text{Bank Assets} \div \text{Bank Capital})$$

Key Points

- 1. Liability Sources:** Bank apne funds checkable deposits, nontransactions deposits, aur borrowings se hasil karta hai.
- 2. Bank Capital Role:** Bank capital sudden asset value drop ya unexpected withdrawals ke against cushion ka kaam karta hai.
- 3. ROA and ROE Relationship:** ROE, ROA ko leverage ratio (bank assets/bank capital) se multiply karke nikalta hai.
- 4. Profitability Measure:** Net interest income bank ke receive kiye gaye aur diye gaye interest ka farq hota hai.
- 5. Off-Balance-Sheet Activities:** Banks fee income generate karne ke liye lines of credit aur letters of credit jaisi activities karte hain, jo risk bhi create karti hain.

Superstructure

- 1. Liabilities:** Checkable deposits, nontransactions deposits, aur borrowings.
- 2. Bank Capital:** Owners ka stake, jisme loan loss reserves bhi shamil hain.
- 3. Profitability Measures:** ROA, ROE, aur net interest margin.
- 4. Off-Balance-Sheet Items:** Lines of credit aur letters of credit.

Example

Socho tumhare dost ne tumse waada kiya ke woh apni dukaan ke liye saman kharidega aur baad mein supplier ko payment kar dega, magar supplier ko yaqeen nahi hai. Agar koi trusted banda (jaise bank) guarantee de de ke "agar mera dost paisay na de sake tou main de dunga", tou supplier asaani se saman de dega. Yahi letter of credit ki tarah kaam karta hai jahan bank apni guarantee

Conceptual Questions

Q: What is the relationship between Return on Assets (ROA) and Return on Equity (ROE), and why does it matter for understanding bank leverage?

Answer: ROE, ROA ko bank assets aur bank capital ke ratio (leverage) se multiply karke nikalta hai, isliye agar bank zyada leverage use kare tou ROA kam hone ke bawajood ROE zyada ho sakta hai.

Jaise table mein dekho, 1996 mein ROA sirf 0.0175 tha magar ROE 0.2147 tha, kyunke bank apne capital se kahin zyada assets manage kar raha tha.

Q: Why do banks engage in off-balance-sheet activities like letters of credit, and what risk does this create?

Answer: Banks yeh activities fee income kamane ke liye karte hain, jahan woh apni guarantee customer ki jagah deta hai taake transaction asaani se ho sake.

Jaise agar koi importer kisi exporter se saman kharide aur payment ka risk ho, tou bank letter of credit issue kar deta hai jisse transaction ho jati hai, magar agar importer default kare tou bank ko risk uthana parta hai.

LESSON QUIZ

1. What percentage of commercial bank liabilities do nontransactions deposits account for? (**Nearly two-thirds**)
2. What is a repurchase agreement (repo)? (**A short-term collateralized loan**)
3. What formula is used to calculate Return on Assets (ROA)? (**Net profit after taxes divided by total bank assets**)
4. What is the difference between interest received and interest paid by a bank called? (**Net interest income**)
5. What off-balance-sheet activity guarantees a customer's promised payment? (**Letter of credit**)

LESSON 25

Bank Risk

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apni motorbike kisi ko thori der ke liye lend kar do, magar agar achanak tumhe usi waqt zaroorat par jaye aur woh wapas na mil sake, tou tumhe mushkil ho jati hai. Banks ko bhi waisi hi mushkil hoti hai jab depositors achanak apna paisa wapas maangte hain ya interest rates badal jate hain.

Key Terms

1. **Liquidity Risk:** Yeh risk hai jab bank ko achanak bohot zyada funds ki demand pori karni paray.
2. **Excess Reserves:** Yeh required reserves se zyada cash hai jo bank withdrawal requests handle karne ke liye rakhta hai.
3. **Asset Management:** Yeh liquidity risk handle karne ka tareeqa hai jisme bank securities ya loans bech kar paisa hasil karta hai.
4. **Liability Management:** Yeh tareeqa hai jisme bank borrowing ya naye deposits attract karke paisa hasil karta hai.
5. **Credit Risk:** Yeh risk hai ke loans wapas nahi milenge.
6. **Interest-Rate Risk:** Yeh risk hota hai jab bank ki assets aur liabilities interest rate changes se mukhtalif tarah se affect hoti hain.
7. **Net Interest Margin:** Yeh assets se milne wala interest minus liabilities par diya gaya interest hai.

8. Gap Analysis: Yeh interest-rate sensitive assets aur liabilities ke farq ko measure karta hai.

Key Points

- 1. Liquidity Risk Source:** Yeh risk deposit withdrawals aur off-balance sheet activities, dono taraf se aa sakta hai.
- 2. Excess Reserves Cost:** Excess reserves rakhna expensive hota hai kyunke us paisay par interest nahi milta.
- 3. Two Adjustment Methods:** Liquidity risk asset adjustment (securities/loans bechna) ya liability adjustment (borrowing/deposits attract karna) se manage hoti hai.
- 4. Credit Risk Management:** Yeh diversification, screening, monitoring, aur collateral demand karne se kam ki ja sakti hai.
- 5. Interest-Rate Risk Impact:** Agar bank ki liabilities assets se zyada interest-rate sensitive hon, tou interest rate barhne se bank ka profit kam ho jata hai (negative gap).

Item	Assets	Liabilities
Interest rate sensitive	\$20	\$50
Not interest rate sensitive	\$80	\$50

Example

Socho tumhare pass ek chai ki dukaan hai jahan tumne kuch paisa cash mein rakha hai (reserves) aur kuch logon ko udhaar diya hai (loans). Agar achanak tumhara koi customer apna pura udhaar wapas maange aur tumhare pass cash kam ho, tou tumhe apne kisi asset ko bechna paray ga ya kisi se udhaar lena paray ga. Yahi liquidity risk hai jo banks ko face karna parta hai, sirf bohot bare scale par.

Conceptual Questions

Q: What is the difference between managing liquidity risk through asset adjustment versus liability adjustment?

Answer: Asset adjustment mein bank apni securities ya loans bech kar paisa hasil karta hai, jiski waja se bank ka size shrink ho jata hai, jabke liability adjustment mein bank borrowing ya naye deposits attract karke paisa hasil karta hai, jisse bank ka size waisa hi rehta hai.

Jaise table mein dekho, jab withdrawal securities bech kar pora kiya gaya tou total assets kam ho gaye, magar jab borrowing se pora kiya gaya tou assets same rahe sirf liabilities ka composition change hua.

Q: Why does a negative gap (more interest-rate sensitive liabilities than assets) hurt a bank's profits when interest rates rise?

Answer: Jab interest rate barhta hai, tou bank ki sensitive liabilities ka cost zyada barh jata hai bajaye uski sensitive assets ke revenue ke, isliye net profit kam ho jata hai.

Jaise example mein dekho, interest rate 1% barhne se revenue \$5.00 se \$5.20 hua (sirf \$0.20 increase), magar cost \$3.00 se \$3.50 ho gaya (\$0.50 increase), isliye profit \$2.00 se \$1.70 reh gaya.

LESSON QUIZ

- 1. What is liquidity risk? **(The risk of a sudden demand for funds that a bank cannot meet)**
- 2. Name the two ways a bank can manage liquidity risk on its balance sheet. **(Adjusting assets and adjusting liabilities)**
- 3. What is credit risk? **(The risk that loans will not be repaid)**
- 4. Why do bank balance sheets typically face interest-rate risk? **(Because liabilities tend to be short-term while assets tend to be long-term)**
- 5. What does a negative gap between interest-rate sensitive assets and liabilities indicate? **(That a rise in interest rates will reduce the bank's profits)**

LESSON 26

Interest Rate Risk (Cont.) and Globalization of Banking

Deep Explanation

Yeh bilkul waisi baat hai jaise koi business sirf apne sheher tak mehdood na rahe, balke poori dunya mein customers dhoondhe aur har jagah ke local risks (jaise currency ka rate, government ki policy) ko bhi samjhe. Banks bhi jab international level par operate karte hain, tou unhe nayi tarah ke risks face karne parte hain.

Key Terms

- 1. Gap Analysis:** Yeh interest-sensitive assets aur liabilities ke yield ka farq batata hai jo profit change predict karta hai.
- 2. Trading Risk:** Yeh risk hai jo banks ko securities, loans, aur derivatives trade karte waqt face hota hai.
- 3. Foreign Exchange Risk:** Yeh risk currency exchange rate ke unfavorable movement se ata hai.
- 4. Sovereign Risk:** Yeh risk hota hai jab koi government loan repayment ko prohibit kar de.
- 5. Operational Risk:** Yeh risk hai jo computer systems ya buildings ke fail ho jane se ata hai.
- 6. International Banking Facility (IBF):** Yeh facility foreigners se deposits accept karti hai aur unhe loans deti hai, mulk ke bahar.
- 7. Edge Act Subsidiary:** Yeh subsidiary international banking transactions mein engage karti hai.
- 8. Eurodollar Market:** Yeh market hai jahan dollar-denominated deposits foreign banks mein exchange hoti hain.
- 9. LIBOR:** Yeh woh interest rate hai jis par banks ek dusre ko Eurodollars lend karte hain.
- 10. Financial Holding Company:** Yeh universal bank ki limited form hai jo banking, insurance, aur securities sab kuch karti hai.

Key Points

- 1. Interest-Rate Risk Management:** Banks asset aur liability ki interest-rate sensitivity ko match karke yeh risk kam karte hain, magar yeh credit risk barha sakta hai.
- 2. Trading Risk:** Banks traders par limits lagate hain aur zyada risk hone par zyada capital rakhte hain.
- 3. International Risks:** Foreign banks ko foreign exchange risk aur sovereign risk dono ka samna karna parta hai.
- 4. Globalization Impact:** International banking se competition aur efficiency barhi hai, magar profits kamana mushkil ho gaya hai.
- 5. Future Trends:** Banking industry mein do opposite trends hain — bari companies one-stop shopping dena chahti hain, aur saath mein chote firms specific services provide karte hain.

Example

Socho ek Pakistani student foreign university mein admission ke liye loan lena chahta hai. Agar woh sirf apne sheher ke bank tak mehdood rahe tou usay limited options milengi, magar agar woh international banks se bhi shop kar sake tou usay behtar rates mil sakti hain. Yahi globalization of banking ki misal hai, jahan borrowers aur depositors dono ko duniya bhar mein options milti hain.

Conceptual Questions

Q: Why does matching the interest-rate sensitivity of assets and liabilities reduce interest-rate risk but increase credit risk?

Answer: Jab bank short-term securities khareed kar variable rate deposits ko match karta hai ya floating rate par long-term loans deta hai, tou interest-rate risk kam ho jata hai, magar floating rate loans borrowers ke liye zyada risky hote hain jo unke default karne ke chances barha dete hain.

Jaise agar bank kisi ko floating rate par ghar ka loan de aur interest rate barh jaye, tou borrower ki monthly payment bhi barh jayegi jo uske default karne ka risk barha degi.

Q: What is the difference between foreign exchange risk and sovereign risk in international banking?

Answer: Foreign exchange risk currency ke exchange rate mein unfavorable change se ata hai, jabke sovereign risk us waqt ata hai jab kisi mulk ki government loan repayment ko rok de.

Jaise agar bank ne kisi foreign company ko loan diya ho aur uska currency value gir jaye, tou yeh foreign exchange risk hai; magar agar us mulk ki government hi loan repayment ko ban kar de, tou yeh sovereign risk hai.

LESSON QUIZ

1. What does gap analysis measure? **(The difference between the yield on interest-sensitive assets and interest-sensitive liabilities)**
2. What is trading risk also known as? **(Market risk)**
3. Name one way banks manage foreign exchange risk. **(Attracting deposits in the same currency as loans, or using foreign exchange futures/swaps)**
4. What is the Eurodollar market? **(A market where dollar-denominated deposits in foreign banks are exchanged)**
5. What interest rate serves as the standard for many private loan rates based on interbank Eurodollar lending? **(LIBOR)**

LESSON 27

Non-Depository Institutions

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apni bike ke liye insurance leta ho — har mahine thora paisa dete ho, aur agar accident ho jaye tou company tumhara nuqsan cover kar deti hai. Insurance companies bhi yeh hi kaam karti hain, sirf bohot bare scale par aur kayi tarah ki cheezon ke liye.

Key Terms

1. **Underwriting:** Yeh process hai jisme risk experts mil kar ek syndicate banate hain aur risk ka assessment karte hain.
2. **Premium:** Yeh woh paisa hai jo policyholder insurance company ko deta hai protection ke badle.
3. **Term Life Insurance:** Yeh insurance hai jo policyholder ki death par beneficiaries ko payment deta hai.
4. **Whole Life Insurance:** Yeh term insurance aur savings account ka combination hai.
5. **Property and Casualty Insurance:** Yeh insurance property aur casualty (jaise driver) dono ko cover karta hai.
6. **Brokerage Firm:** Yeh accounting aur secondary markets tak access provide karti hai.
7. **Investment Bank:** Yeh securities firm hai jiska kaam information produce karna hota hai.
8. **Mutual Fund Company:** Yeh bhi securities firms ka hissa hai jo financial intermediary ka kaam karti hai.

Key Points

1. **Risk Transfer:** Insurance individuals ke liye risk transfer karne ka tareeqa hai.
2. **Pooling Function:** Insurance companies choti policies pool karke bari investments karti hain aur risk ko bari population mein diversify karti hain.

3. Two Insurance Types: Life insurance aur property/casualty insurance, yeh do basic types hain.

4. Adverse Selection Severity: Insurance market mein adverse selection aur moral hazard stock/bond markets se bhi zyada problematic hote hain.

5. Screening Methods: Insurance companies medical exams aur driving records dekh kar applicants ko screen karti hain.

6. Securities Firms Role: Brokerage firms, investment banks, aur mutual fund companies sab financial intermediaries ki tarah kaam karti hain.

Types

1. Life Insurance: Term life aur whole life insurance.

2. Property and Casualty Insurance: Jaise auto insurance, jo property aur driver dono ko cover karta hai.

Example

Socho tumhare ghar mein fire extinguisher nahi hai aur tum fire insurance lena chahte ho. Insurance company tumse poochegi ke ghar mein safety measures hain ya nahi, aur agar nahi hain tou shayad woh policy na de ya zyada premium maange. Yahi underwriting aur screening process hai jisse company apna risk kam karti hai, bilkul waise jaise bank loan dene se pehle creditworthiness check karta hai.

Conceptual Questions

Q: Why is the problem of adverse selection and moral hazard considered worse in insurance markets compared to stock and bond markets?

Answer: Insurance mein policyholder ko apni health ya risk ke baare mein zyada information hoti hai jo company ko nahi hoti, jaise ek cancer patient ya kisi ka ghar fire-prone ho, isliye companies ko careful screening aur covenants ka sahara lena parta hai.

Jaise agar koi cancer patient life insurance lena chahe, tou company ko medical examination zaroori hota hai taake woh apna risk sahi tarah assess kar sake, warna sirf high-risk log hi insurance lenge.

Q: What is the key difference between term life insurance and whole life insurance?

Answer: Term life insurance sirf death benefit deta hai, jabke whole life insurance death benefit ke saath ek savings account bhi combine karta hai jiska cash value refund ho sakta hai.

Jaise agar koi sirf apni family ko protect karna chahta hai limited time ke liye, tou term insurance behtar hai, magar agar woh saving bhi karna chahta hai life insurance ke saath, tou whole life insurance zyada appropriate hoga.

LESSON QUIZ

1. What is underwriting in the context of insurance? **(Risk assessment and loss reimbursement guarantee by a syndicate of risk experts)**
2. Name the two main types of insurance companies. **(Life insurance and property and casualty insurance)**
3. What combination makes up auto insurance? **(Property insurance on the car and casualty insurance on the driver)**
4. What is the primary service securities firms provide that is described as being at the heart of investment banking? **(Producing information)**
5. Name two ways insurance companies screen applicants to reduce adverse selection. **(Medical examination and driving records)**

Non-Depository Institutions (Continued) and Banking Crisis

Deep Explanation

Yeh bilkul waisi baat hai jaise koi shaadi ka function organize karne wala banda hota hai jo guests aur caterer ke darmiyan deal karwata hai, aur agar woh trustworthy hoga tou log usse asaani se kaam le lenge. Investment banks bhi yeh hi kaam karte hain — companies aur investors ke darmiyan deal karwana, jahan unki reputation bohot important hoti hai.

Key Terms

- 1. Underwriting (Investment Banking):** Yeh process hai jisme investment bank ek nayi security ki price guarantee karta hai aur usay investors ko zyada price par bechta hai.
- 2. Finance Company:** Yeh commercial paper aur securities issue karke funds raise karti hai aur loans deti hai.
- 3. Government-Sponsored Enterprise:** Yeh government ki taraf se chartered institution hoti hai jo specific financing provide karti hai.
- 4. Bank Run:** Yeh tab hota hai jab depositors achanak apna paisa cash mein convert karne ke liye bank ki taraf bhaagte hain.
- 5. Solvency:** Yeh batata hai ke bank ke assets uski liabilities se zyada hain (positive net worth).
- 6. Liquidity:** Yeh batata hai ke bank ke pass withdrawal demands pori karne ke liye kaafi reserves hain ya nahi.
- 7. Contagion:** Yeh ek bank ki failure ka asar system-wide panic mein badal jana hai.

Key Points

- 1. Investment Bank Risk:** Underwriting mein risk hota hai kyunke selling price guaranteed price se kam bhi ho sakti hai.
- 2. Finance Company Loans:** Yeh consumer, business, aur sales loans mein specialize karti hain.
- 3. Pakistani GSEs:** ZTBL, SME Bank, HBFC, aur Khushhali Bank yeh sab government-sponsored enterprises ki misalein hain.
- 4. Bank Fragility:** Banks fragile hote hain kyunke woh depositors ko on-demand withdrawal ka promise karte hain, first-come first-served basis par.
- 5. Solvency vs Liquidity:** Bank run ke waqt asal masla yeh nahi hota ke bank solvent hai ya nahi, balke yeh hota hai ke woh liquid hai ya nahi.
- 6. Recession Impact:** Recession mein log apni jobs khote hain aur loans pay nahi kar sakte, jis se banks ke assets ki value gir jati hai aur bank failures barh jate hain.

Example

Socho ek bank ke baare mein afwah phail jati hai ke uske pass paisay khatam ho gaye hain, halanke asal mein bank ke assets uski liabilities se zyada hain (yani woh solvent hai). Lekin yeh sun kar sab depositors ek sath apna paisa nikalne ke liye bank pohanch jate hain, aur bank itna cash itni jaldi nahi de sakta, isliye woh "illiquid" ho jata hai aur fail kar jata hai — yahi ek classic bank run hai.

Conceptual Questions

Q: What is the key difference between a bank being insolvent and a bank being illiquid, and why does this distinction matter during a bank run?

Answer: Insolvent hone ka matlab hai ke bank ke liabilities uski assets se zyada hain, jabke illiquid hone ka matlab hai ke bank ke pass abhi withdrawal demands pori karne ke liye kaafi cash nahi hai, chahe woh solvent hi ho.

Jaise ek bank solvent ho sakta hai (uska net worth positive hai) magar agar achanak sab depositors paisay nikalne aa jayen, tou cash ki kami se woh illiquid ho jayega aur fail kar sakta hai, chahe uski underlying financial health acchi ho.

Q: Why are finance companies and investment banks both considered financial intermediaries despite providing very

different services?

Answer: Dono hi transactions aur information costs ko kam karte hain — finance companies loans dene mein specialize karti hain jabke investment banks securities issue karne aur firms ko advice dene mein.

Jaise finance company kisi consumer ko appliance khareedne ke liye loan degi, jabke investment bank kisi company ko stock market mein nayi shares issue karne mein madad degi — dono apne apne tareeqe se funds ko savers se borrowers tak pohanchate hain.

LESSON QUIZ

1. What service do investment banks provide when they guarantee the price of a new security issue? (**Underwriting**)
2. Name the three main loan types finance companies specialize in. (**Consumer loans, business loans, and sales loans**)
3. What is the term for a single bank failure spreading into a system-wide panic? (**Contagion**)
4. What is the difference between solvency and liquidity for a bank? (**Solvency means assets exceed liabilities; liquidity means having sufficient reserves to meet withdrawals**)
5. What economic event historically increases the risk of bank panics by reducing borrowers' ability to repay loans? (**A recession**)

LESSON 29

The Government Safety Net

Deep Explanation

Yeh bilkul waisi baat hai jaise koi senior tumhe yaqeen dilata hai ke agar tum kisi naye shop se saman khareedo aur woh fraud nikle, tou koi authority tumhari madad karegi. Government bhi financial system mein logon ka paisa protect karne ke liye yeh hi role play karti hai.

Key Terms

1. **Investor Protection:** Yeh government ka role hai jisme small investors ko financial institutions ki mismanagement se bachaya jata hai.
2. **Lender of Last Resort:** Yeh government ka role hai jisme woh solvent banks ko unke depositors ki withdrawal demands pori karne mein madad deti hai.
3. **Moral Hazard (Safety Net):** Yeh problem hai jab depositors apne bank ko monitor nahi karte kyunke unhe pata hota hai ke unka paisa protected hai, isliye banks zyada risk lete hain.
4. **Too-Big-to-Fail:** Yeh policy hai jisme bohot bari institutions ko pata hota hai ke government unhe bail out karegi agar woh fail hone lagein.
5. **Regulation:** Yeh rules hain jo bank managers ko follow karni hoti hain.
6. **Supervision:** Yeh financial institutions ki general oversight hai.
7. **CAMELS:** Yeh acronym hai jo bank ki health evaluate karne ke criteria batata hai — Capital adequacy, Asset quality, Management, Earnings, Liquidity, Sensitivity to risk.

Key Points

1. **Three Government Reasons:** Investors ko protect karna, monopolistic exploitation se bachana, aur financial system ki stability ensure karna.
2. **Depository Institutions Importance:** Banks ko zyada attention milti hai kyunke woh payments system mein central role rakhte hain aur runs ka shikar ho sakte hain.

3. Illiquid vs Insolvent Distinction: Central bank officials ko illiquid aur insolvent institution mein farq karna zaroori hota hai loan dene se pehle.

4. Safety Net Problems: Government protection moral hazard create karti hai kyunke depositors apne bank ko monitor nahi karte.

5. Pakistan's Regulator: State Bank of Pakistan (SBP) Pakistan mein banking sector ki supreme regulatory authority hai.

Example

Socho tumhare mohalle mein ek dukaan hai jise sab jante hain ke "bohot bari hai, yeh kabhi band nahi hogi", isliye log uske saman ki quality check karna chor dete hain. Lekin agar dukaandar ko bhi yeh pata ho ke uski dukaan kabhi fail nahi hone degi (jaise koi bara investor usay bachata rahega), tou woh bhi laparwahi karna shuru kar dega. Yahi too-big-to-fail problem hai jo banks ke sath hoti hai.

Conceptual Questions

Q: Why does the government safety net create a moral hazard problem instead of simply solving the bank run problem?

Answer: Jab depositors ko pata hota hai ke unka paisa government ki taraf se protected hai, tou woh apne bank ke risky behavior ko monitor karna chor dete hain, jisse banks zyada risk lene lagte hain kyunke unhe pata hota hai ke consequences kam honge.

Jaise too-big-to-fail banks ke managers jante hain ke agar woh fail honay lagein tou government bail out karegi, isliye woh normal se zyada risk le sakte hain.

Q: Why is it important for central bank officials to distinguish between an illiquid bank and an insolvent bank before approving a loan?

Answer: Agar bank sirf illiquid hai (yani temporarily cash ki kami hai magar uska net worth positive hai) tou loan dena sahi hai aur panic ko rok sakta hai, magar agar bank insolvent hai (uski liabilities assets se zyada hain) tou loan dena sirf taxpayers ka paisa zaya karega.

Jaise agar ek healthy bank sirf rumors ki waja se withdrawal pressure mein hai, tou usay loan dena system ko stable rakhega, magar ek bohot kharab financial health wale bank ko bachana sahi nahi hoga.

LESSON QUIZ

1. Name the three reasons for government involvement in the financial system. **(Protecting investors, protecting customers from monopolistic exploitation, and ensuring financial system stability)**
2. What does "lender of last resort" mean? **(The government/central bank provides loans to solvent banks during financial stress)**
3. What problem does protecting depositors create for banks' risk-taking behavior? **(Moral hazard)**
4. What does the acronym CAMELS stand for in bank evaluation? **(Capital adequacy, Asset quality, Management, Earnings, Liquidity, Sensitivity to risk)**
5. Which institution is the supreme regulatory authority for banking in Pakistan? **(State Bank of Pakistan, SBP)**

Yeh bilkul waisi baat hai jaise ek ghar ka head apne ghar ke budget ko control karta hai taake na zyada kharcha ho na bohot kam, balke sab kuch stable rahe. Central bank bhi poori economy ke liye yeh hi kaam karta hai — money supply aur interest rates ko control kar ke stability laata hai.

Key Terms

- 1. Central Bank:** Yeh institution currency issue karne ka monopoly rakhta hai aur money/credit ki availability control karta hai.
- 2. Monetary Policy:** Yeh short-term interest rates ko adjust karne ki activity hai.
- 3. Expansionary Policy:** Yeh policy interest rates kam kar ke growth aur inflation ko barhati hai.
- 4. Restrictive Policy:** Yeh policy interest rates barha kar growth aur inflation ko kam karti hai.
- 5. Price Stability:** Yeh central bank ka woh goal hai jisme inflation ko control mein rakha jata hai.
- 6. Deflation:** Yeh prices ka girna hai, jo loan defaults aur banking problems ko barha sakta hai.

Key Points

- 1. Bankers' Bank Role:** Central bank lender of last resort ka kaam karta hai, payments system manage karta hai, aur commercial banks ko oversee karta hai.
- 2. Central Bank Limitations:** Central bank securities markets aur government ke budget ko control nahi karta.
- 3. Five Stability Objectives:** Low inflation, high growth/employment, stable financial markets, stable interest rates, aur stable exchange rate.
- 4. Tradeoffs Necessary:** Sab paanch objectives ko ek sath achieve karna mumkin nahi hota, isliye tradeoffs karne parte hain.
- 5. Zero Inflation Risk:** Zero inflation deflation ka risk barhata hai, jo loan defaults aur bank health ko nuqsan pohancha sakta hai.

Example

Socho ek dukaandar apne dukaan ke rates itne barha de ke log saman khareedna chor dein, ya itne kam kar de ke usay nuqsan ho jaye. Usay ek balance dhoondhna parta hai. Central bank bhi yeh hi kaam karta hai poori economy ke liye — interest rates ko itna control karta hai ke na zyada inflation ho na economy ruk jaye.

Conceptual Questions

Q: Why do central banks avoid targeting zero inflation as their goal, even though they want price stability?

Answer: Zero inflation deflation ka risk barhata hai, jisse loan defaults barh jate hain aur banks ki health kharab ho sakti hai, aur employers ke liye nominal wages kam karna mushkil ho jata hai.

Jaise agar prices girna shuru ho jayen, tou log apne loans pay karne mein mushkil feel karenge kyunke unki income ki value bhi gir rahi hogi, jo banks ke liye risky hai.

Q: What is the key difference between expansionary and restrictive monetary policy, and when would a central bank use each?

Answer: Expansionary policy mein interest rates kam kiye jate hain taake growth aur inflation barhe, jabke restrictive policy mein interest rates barhaye jate hain taake growth aur inflation kam ho.

Jaise agar economy slow ho rahi ho aur unemployment barh rahi ho, tou central bank expansionary policy apnayega, magar agar inflation bohot zyada barh rahi ho, tou restrictive policy use karega.

LESSON QUIZ

1. What monopoly does the central bank hold? **(The issuance of currency)**
2. What is the term for adjusting short-term interest rates to influence the economy? **(Monetary policy)**
3. Name two things a central bank does NOT control. **(Securities markets and the government's budget)**

4. List the five objectives central bankers pursue for stability. (**Low and stable inflation, high and stable real growth/employment, stable financial markets, stable interest rates, stable exchange rate**)
5. Why is zero inflation considered too low a target? (**It risks deflation, which increases loan defaults and threatens bank health**)

LESSON 31

Low, Stable Inflation and Central Bank Objectives

Deep Explanation

Yeh bilkul waisi baat hai jaise ek teacher apni class mein discipline rakhta hai — na bohot zyada strict ke students dab jayen, na bohot loose ke sab kuch bigar jaye, balke ek balance rakhta hai. Central bank bhi economy mein yeh hi balance rakhne ki koshish karta hai, taake na bohot zyada inflation ho na deflation.

Key Points

- 1. Price Stability Goal:** Central banks inflation ko low aur stable rakhna chahte hain taake money ki value predictable rahe.
- 2. Zero Inflation Risk:** Zero inflation deflation ka khata barhata hai, jo loan defaults aur banking problems create karta hai.
- 3. Nominal Wage Rigidity:** Agar inflation zero ho tou employers ke liye nominal wages kam karna mushkil ho jata hai, isliye thori inflation labor market ko behtar chalati hai.
- 4. Growth Stabilization:** Central bankers interest rates adjust kar ke business cycle ki fluctuations ko kam karte hain.
- 5. Interest Rate Volatility Problems:** Yeh output ko unstable banati hai aur risk premium ko barhati hai, jisse financial decisions mushkil ho jate hain.
- 6. Exchange Rate Priority:** Exchange rate stability central bank ki priority list mein sabse aakhir mein ati hai, magar developing countries ke liye yeh zyada important hoti hai.

Key Terms

- 1. Price Stability:** Yeh inflation ko low aur predictable rakhne ka goal hai.
- 2. Deflation:** Yeh prices ka girna hai jo loan defaults aur bank health ko nuqsan deta hai.
- 3. Business Cycle:** Yeh economy ke boom aur recession ke fluctuations hain.
- 4. Interest Rate Stability:** Yeh borrowing aur expenditure ko predictable rakhne ka tareeqa hai.
- 5. Exchange Rate Stability:** Yeh imports aur exports ki prices ko predictable rakhne ka tareeqa hai.
- 6. Central Bank Independence:** Yeh political pressure se azaad hona hai, jo successful central banking ke liye zaroori hai.

Types

- 1. Modern Central Bank Objectives:** Low stable inflation, high stable growth, financial system stability, stable interest rates, stable exchange rates.
- 2. Successful Central Bank Qualities:** Independent, accountable, transparent, aur clear in communication.

Example

Socho ek dukaandar apne saman ki prices har hafte change karta rahe — kabhi bohot barha de, kabhi bohot kam kar de. Customers confuse ho jayenge ke ab future mein kya price hogi, aur woh apna budget plan nahi kar sakenge. Yahi waja hai ke stable aur predictable prices (low inflation) economy ke liye zaroori hain, taake log aur businesses asaani se planning kar sakein.

Conceptual Questions

Q: Why would a small amount of inflation be considered better for the labor market than zero inflation?

Answer: Agar inflation zero ho tou employers ke liye nominal wages kam karna mushkil hota hai, kyunke log apni salary mein cut dekhna pasand nahi karte, magar thori inflation ki waja se real wages automatically adjust ho jate hain bina nominal wage kam kiye.

Jaise agar kisi employee ki salary same rahe magar inflation 2% ho, tou uski real purchasing power thori kam ho jati hai, jo employer ke liye labor cost adjust karne ka asaan tareeqa hai bina seedha wage cut kiye.

Q: Why is exchange rate stability considered more important for developing countries than for developed ones?

Answer: Developing countries ki economy zyada tar imports aur exports par depend karti hai, isliye exchange rate mein fluctuation unki revenues aur costs ko unpredictable bana deta hai.

Jaise agar Pakistan ki currency achanak gir jaye, tou imported items (jaise oil ya machinery) bohot expensive ho jayenge, jo poori economy ko affect karega.

LESSON QUIZ

1. Why is zero inflation considered risky for an economy? **(It increases the risk of deflation, leading to loan defaults and threats to bank health)**
2. How do central bankers typically moderate recessions? **(By lowering interest rates)**
3. Why is interest rate volatility considered a problem? **(It makes output unstable and increases risk premiums)**
4. Which central bank objective is typically last on the priority list? **(Stable exchange rates)**
5. Name two qualities a successful modern central bank must have. **(Independent of political pressure, accountable to the public, transparent in policy actions, or clear in communications)**

LESSON 32

Creating a Successful Central Bank

Deep Explanation

Yeh bilkul waisi baat hai jaise ek university ka exam department independent hona chahiye taake koi bhi teacher ya student apni marzi se result change na kar sake, magar saath mein woh department apna kaam transparently bhi karta hai taake sab ko pata ho ke results kaise nikalte hain. Central bank ko bhi independent hona chahiye magar saath mein accountable aur transparent bhi.

Key Terms

1. **Central Bank Independence:** Yeh do parts par mabni hai — apna budget control karna aur outside logon se policies reverse na honay dena.
2. **Decision-Making by Committee:** Yeh tareeqa hai jisme group decisions individual decisions se behtar hoti hain.
3. **Accountability:** Yeh policymakers ka public ke saamne apne goals ke progress report karna hai.
4. **Transparency:** Yeh disclosure requirements hain jo policymakers ki clarity barhati hain.
5. **Fiscal Policy:** Yeh taxes aur spending ke baare mein decisions hain jo elected officials karte hain.
6. **Discount Loans:** Yeh central bank ke loans hain jo commercial banks ko short-term cash ki zaroorat par diye jate hain.

7. **Float:** Yeh check-clearing process ka byproduct hai jab central bank ek bank ka account credit kar deta hai dusre ko debit karne se pehle.

Key Points

1. **Four Central Bank Qualities:** Independent, accountable, transparent, aur clear communication.
2. **Committee Decisions Better:** Group ki knowledge aur experience individual quirks ke risk ko kam karti hai.
3. **Political Pressure Risk:** Politicians short-term goals ke liye low interest rates chahte hain, jo long-term inflation create kar sakta hai.
4. **Fiscal-Monetary Conflict:** Fiscal policymakers borrowing aur inflation ko apne funding ke liye use kar sakte hain, jo monetary policy ke goals se takra sakta hai.
5. **Balance Sheet Transparency:** Central bank apna balance sheet regularly publish karta hai jo transparency ka hissa hai.

Superstructure

1. **Assets:** Securities, foreign exchange reserves, aur loans (discount loans aur float).
2. **Liabilities:** Currency, government's deposit account, aur commercial banks ke reserves.

Example

Socho ek class monitor ko teacher har baat mein interfere kar ke uska decision change kar de, tou monitor kabhi bhi sahi tareeqe se kaam nahi kar sakega. Lekin agar monitor ko azaadi mile apne decisions lene ki, magar woh sab kuch class ko transparently bataye, tou yeh ek behtar system hoga. Yahi central bank independence aur accountability ka concept hai.

Conceptual Questions

Q: Why is central bank independence important for controlling inflation, even though it seems to conflict with democratic accountability?

Answer: Politicians short-term mein low interest rates aur high money growth pasand karte hain taake output aur employment barhe, magar yeh long-term mein inflation create karta hai, isliye central bank ko political pressure se azaad rakha jata hai taake woh long-term stability par focus kar sake.

Jaise agar election se pehle politicians interest rates kam karwa dein taake economy boom kare, tou yeh short-term mein achi lage gi magar inflation ka khatra barh jayega — isliye independence zaroori hai.

Q: What is the difference between the central bank's role as the "government's bank" and its role as the "bankers' bank"?

Answer: Government's bank ke role mein central bank securities aur foreign exchange reserves rakhta hai, aur government ka account manage karta hai; bankers' bank ke role mein woh commercial banks ko loans deta hai aur unke reserves rakhta hai.

Jaise government's bank role mein central bank tax revenues ka account rakhta hai, jabke bankers' bank role mein woh commercial banks ko discount loans deta hai jab unhe short-term cash ki zaroorat ho.

LESSON QUIZ

1. What are the two components of central bank independence? (**Control of its own budget and policies not reversible by outsiders**)
2. Why is decision-making by committee preferred over decision-making by an individual? (**It reduces the risk of policy being dictated by an individual's quirks**)
3. Who is responsible for fiscal policy decisions? (**Elected officials**)
4. What percentage of a central bank's liabilities does currency typically account for? (**Over 90 percent**)
5. Name the three basic assets on a central bank's balance sheet. (**Securities, foreign exchange reserves, and loans**)

The Monetary Base

Deep Explanation

Yeh bilkul waisi baat hai jaise tumhare ghar mein total cash sirf tab change hota hai jab koi bahar se paisa aaye ya bahar jaye, magar agar tum apne hi ghar mein wallet se almari mein paisa shift karo tou total amount same rehta hai. Monetary base bhi waisi hi hai — kuch transactions uska size change karti hain, aur kuch sirf uski composition.

Key Terms

- 1. Monetary Base:** Yeh currency in hands of public aur banking system ke reserves ka combination hai, jise high-powered money bhi kehte hain.
- 2. Bank Reserves:** Yeh vault cash plus central bank mein deposits ka majmua hai.
- 3. Open Market Operation (OMO):** Yeh transaction hai jisme central bank securities khareedta ya bechta hai.
- 4. Foreign Exchange Intervention:** Yeh transaction hai jisme central bank foreign currency reserves khareedta ya bechta hai.
- 5. Discount Loan:** Yeh loan hai jo central bank commercial banks ko deta hai jab unhe short-term cash chahiye ho.
- 6. Cash Withdrawal:** Yeh action hai jab koi individual apne bank account se cash nikalta hai.

Key Points

- 1. Balance Sheet Rule:** Agar koi asset barhe tou koi dusra asset kam hoga ya liability barhegi, taake net change zero rahe.
- 2. Three Size-Changing Transactions:** Open market operations, foreign exchange interventions, aur discount loans — yeh teen central bank ke balance sheet ka size change karte hain.
- 3. Cash Withdrawal Effect:** Yeh sirf monetary base ki composition change karta hai (reserves se currency mein shift), size nahi.
- 4. OMO Mechanics:** Jab central bank Treasury bond khareedta hai, uske assets aur liabilities (reserves) dono barh jate hain, jo monetary base ko barhata hai.
- 5. Discount Loan Initiation:** Yeh commercial bank ki taraf se request hoti hai, central bank zabardasti loan nahi deta.

Types

- 1. Size-Changing Transactions:** Open market operations, foreign exchange interventions, discount loans.
- 2. Composition-Changing Transactions:** Cash withdrawals.

Example

Socho central bank ek Treasury bond khareedta hai bank se. Bank ko paisa milta hai jo uske reserves mein add ho jata hai, aur central bank ke pass woh bond asset ban jata hai. Total monetary base barh jati hai kyunke naye reserves create hue hain. Lekin agar koi banda apne account se cash nikal kar apni pocket mein rakh le, tou sirf currency aur reserves ke darmiyan shift hua hai, total amount same hai.

Conceptual Questions

Q: Why does an open market purchase increase the monetary base, while a cash withdrawal does not, even though both involve changes to the central bank's balance sheet?

Answer: Open market purchase mein central bank ka total balance sheet size barhta hai kyunke naya asset (security) aur naya liability (reserves) dono create hote hain, magar cash withdrawal mein sirf existing liabilities ke darmiyan shift hota hai (reserves se

currency mein), total size same rehta hai.

Jaise jab central bank bond khareedta hai tou \$1 billion ke reserves naye create hote hain, magar jab koi cash nikalta hai tou sirf \$100 reserves se currency mein move hote hain, koi naya paisa create nahi hota.

Q: Why are discount loans initiated by commercial banks rather than forced by the central bank?

Answer: Discount loans ka system is liye banaya gaya hai ke commercial banks apni zaroorat ke mutabiq short-term cash request kar sakein, aur unhe collateral (jaise Treasury bond) dena parta hai, isliye yeh ek voluntary request hoti hai na ke central bank ki taraf se majboori.

Jaise agar kisi bank ko achanak withdrawal pressure ka samna ho aur uske pass kaafi reserves na hon, tou woh central bank se discount loan request kar sakta hai apna collateral dekha kar.

LESSON QUIZ

1. What are the two components of the monetary base? **(Currency in the hands of the public and bank reserves)**
2. What is another name for the monetary base? **(High-powered money)**
3. Name the three transactions that change the size of the monetary base. **(Open market operations, foreign exchange interventions, discount loans)**
4. What happens to the monetary base when someone withdraws cash from an ATM? **(It stays the same; only the composition shifts)**
5. Who initiates a discount loan request? **(The commercial bank)**

LESSON 34

Deposit Creation in a Single Bank

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apne dost ko paisa do, woh agay kisi dukaandar ko de deta hai, dukaandar woh paisa bank mein jama kar deta hai, aur bank phir us se kuch hissa kisi aur ko loan de deta hai. Yeh chain chalti rehti hai aur ek hi original paisa system mein bohot zyada paisa create kar deta hai.

Key Terms

1. **Actual Reserves (R):** Yeh bank ke pass total reserves hain.
2. **Required Reserves (RR):** Yeh reserves hain jo bank ko regulation ke mutabiq rakhni parti hain ($r_D \times D$).
3. **Excess Reserves (ER):** Yeh reserves hain jo required reserves se zyada hain.
4. **Reserve Requirement Ratio:** Yeh percentage hai jo deposits ka hissa reserves mein rakhna zaroori hota hai.
5. **Deposit Expansion Multiplier:** Yeh $1/r_D$ hota hai, jo batata hai ke reserves mein har dollar increase se deposits kitne barhenge.
6. **Money Multiplier:** Yeh batata hai ke quantity of money monetary base se kitna relate karti hai ($M = m \times MB$).
7. **Excess Reserve Ratio:** Yeh ER/D hota hai, jo banks ki apni marzi se zyada reserves rakhne ki tendency batata hai.

Key Points

1. **Single Bank Lending:** Jab central bank ek bank se security khareedta hai, bank ko excess reserves milti hain jo woh loan dene ke liye use karta hai.
2. **Multiple Deposit Creation:** Ek bank ka loan dusre bank mein deposit ban jata hai, aur yeh chain har bank mein chalti rehti hai jab

tak sara paisa required reserves mein na chala jaye.

3. Deposit Expansion Formula: $\Delta D = (1/r_D) \times \Delta RR$, jahan r_D reserve requirement ratio hai.

4. Excess Reserves and Cash Withdrawal Effect: Agar banks excess reserves rakhein ya log cash withdraw karein, tou deposit expansion ka asar kam ho jata hai.

5. Money Multiplier Relationship: Money (currency + deposits) monetary base (currency + reserves) se multiplier ke through related hai.

Bank	Increase in Deposits	Increase in Loans	Increase in Reserves
First Bank	\$0	\$100,000	\$0
Second Bank	\$100,000	\$90,000	\$10,000
Third Bank	\$90,000	\$81,000	\$9,000
Banking System Total	\$1,000,000	\$1,000,000	\$100,000

Example

Socho First Bank ne OBI ko \$100,000 ka loan diya, jo OBI ne American Steel Co ko payment ke through diya, aur American Steel ne yeh paisa Second Bank mein deposit kar diya. Agar reserve requirement 10% hai, tou Second Bank \$10,000 reserve mein rakh kar baqi \$90,000 kisi aur ko loan de degi, aur yeh process aise hi chalti rahegi har bank mein. Is tarah ek hi \$100,000 ka original paisa system mein \$1,000,000 tak deposits banane mein madad karta hai.

Conceptual Questions

Q: Why does a single \$100,000 open market purchase by the central bank lead to a \$1,000,000 increase in total deposits across the banking system?

Answer: Har bank apna deposit milne ke baad sirf required reserve (10%) rakhta hai aur baqi loan de deta hai, jo agay kisi aur bank mein deposit ban jata hai, aur yeh chain reserves khatam honay tak chalti rehti hai, isliye total effect deposit expansion multiplier ($1/r_D$) ke through barh jata hai.

Jaise table mein dekho, First Bank ne \$100,000 loan diya, Second Bank ko \$100,000 deposit mila jisme se \$90,000 loan diya, Third Bank ko \$90,000 mila jisme se \$81,000 loan diya, aur yeh silsila chalta raha jab tak total \$1,000,000 deposits ban gaye.

Q: How do excess reserves and cash withdrawals affect the deposit expansion process compared to the simple multiplier model?

Answer: Jab banks excess reserves rakhte hain ya log cash withdraw karte hain, tou kam paisa loan dene ke liye available hota hai, isliye deposit expansion ka asar simple multiplier ($1/r_D$) se kam ho jata hai.

Jaise example mein American Steel ne 5% cash withdraw kiya aur bank ne 5% excess reserves rakhi, jiski waja se Second Bank sirf \$80,750 loan de saki bajaye \$90,000 ke, jo expansion process ko slow kar deta hai.

LESSON QUIZ

- What happens to a bank's balance sheet when the central bank buys a security from it? (**Reserves increase and securities decrease by the same amount**)
- What is the formula for the simple deposit expansion multiplier? (**$1/r_D$, where r_D is the required reserve ratio**)
- With a 10% reserve requirement, what is the maximum total deposit expansion from a \$100,000 increase in reserves? (**\$1,000,000**)
- What two factors reduce the impact of the simple deposit expansion multiplier? (**Banks holding excess reserves and cash withdrawals by the public**)

LESSON 35

Money Multiplier and the Monetary Policy Toolbox

Deep Explanation

Yeh bilkul waisi baat hai jaise ek dukaandar ka asal cash kam ho lekin uske udhaar ki waja se market mein bohot zyada paisa circulate ho raha ho. Monetary base sirf ek choti si rakam hoti hai, magar money multiplier ki waja se woh poori economy mein bohot zyada money supply create kar deti hai.

Key Terms

- 1. Money Multiplier (m):** Yeh batata hai ke monetary base se quantity of money kaise nikalti hai ($M = m \times MB$).
- 2. Currency Ratio {C/D}:** Yeh currency aur deposits ka ratio hai jo public ki currency demand batata hai.
- 3. Excess Reserve Ratio {ER/D}:** Yeh banks ki apni marzi se zyada reserves rakhne ki tendency batata hai.
- 4. Federal Funds Rate:** Yeh interest rate hai jis par banks aapas mein overnight reserves lend/borrow karte hain.
- 5. Discount Rate:** Yeh interest rate hai jis par banks central bank se reserves borrow kar sakte hain.
- 6. Target Federal Funds Rate:** Yeh central bank ka primary monetary policy tool hai.

Key Points

- 1. Three Multiplier Components:** Money multiplier currency ratio, required reserve ratio, aur excess reserve ratio teen factors par depend karta hai.
- 2. Monetary Base Uses:** Monetary base teen jaghon mein use hoti hai — required reserves, excess reserves, aur public ke hath mein currency.
- 3. Quantity of Money Factors:** M monetary base, reserve requirements, banks ki excess reserves holding, aur public ki currency demand par depend karta hai.
- 4. Currency Holding's Dual Effect:** Currency holding multiplier ke numerator aur denominator dono ko affect karti hai, isliye uska overall effect seedha clear nahi hota, magar overall yeh money supply ko kam karti hai.
- 5. Three Monetary Policy Tools:** Target federal funds rate, discount rate, aur reserve requirement.

Questions Asked

- 1. What determines the quantity of money in an economy?** Monetary base, reserve requirements, banks ki excess reserve preference, aur public ki currency demand.
- 2. How does the central bank control the federal funds rate?** Open market operations ke through reserves ki quantity ko manipulate kar ke.
- 3. What are the central bank's main policy tools?** Target federal funds rate, discount rate, aur reserve requirement.

Example

Socho ek dukaandar apne pass sirf Rs. 1000 ka cash rakhta hai magar apne customers ko Rs. 10,000 tak ka udhaar (credit) deta hai, kyunke usay pata hai ke sab log ek sath apna paisa wapas nahi maangenge. Yahi tareeqa central bank ke monetary base ke sath hota hai — choti si reserve se banking system bohot zyada money create kar deta hai through loans aur deposits.

Q: Why does an increase in the currency-to-deposit ratio $\{C/D\}$ decrease the quantity of money, even though currency is itself part of the money supply?

Answer: Jab log apna paisa cash mein rakhte hain bajaye bank mein deposit karne ke, tou bank ke pass kam reserves rehti hain jo loan dene ke liye use ho sakti thin, isliye overall money creation process slow ho jata hai aur total money supply kam ho jati hai.

Jaise agar sab log apna paisa cash mein rakhna shuru kar dein, tou banks ke pass loan dene ke liye kam paisa hoga, jisse multiple deposit creation ka process bohot kam ho jayega.

Q: Why does the central bank use open market operations rather than directly setting the federal funds rate by decree?

Answer: Federal funds rate market mein determine hoti hai, isliye central bank seedha order nahi de sakta, balke woh open market operations ke through reserves ki quantity barha ya kam kar ke market rate ko apne target rate tak le ata hai.

Jaise agar central bank chahta hai ke federal funds rate kam ho, tou woh securities khareed kar reserves barha deta hai, jisse banks ke pass zyada paisa hota hai aur woh kam rate par ek dusre ko lend karte hain.

LESSON QUIZ

1. What does the money multiplier (m) relate the quantity of money to? **(The monetary base)**
2. Name the three components that make up the money multiplier formula. **(Currency ratio, required reserve ratio, and excess reserve ratio)**
3. What happens to the quantity of money when the required reserve-to-deposit ratio increases? **(It decreases)**
4. What is the central bank's primary monetary policy instrument? **(The target federal funds rate)**
5. How does the central bank manipulate the federal funds rate? **(Through open market operations, by buying or selling securities to add or drain reserves)**

LESSON 36

Target Federal Funds Rate and Open Market Operations

Deep Explanation

Yeh bilkul waisi baat hai jaise ek school principal apni class ki seats ka supply control karta hai taake "demand" aur "supply" balance rahe — agar zyada students aa jayen tou seats ka rate (priority) barh jata hai. Central bank bhi reserves ki supply ko control kar ke federal funds rate ko apne target tak le ata hai.

Key Terms

1. **Open Market Operations:** Yeh central bank ka tool hai jisme woh securities khareed kar ya bech kar reserves ki quantity ko control karta hai.
2. **Primary Credit:** Yeh sabse short-term loan hai jo sound institutions ko diya jata hai.
3. **Secondary Credit:** Yeh loan hai jo un institutions ko diya jata hai jo primary credit ke liye qualify nahi karte.
4. **Seasonal Credit:** Yeh loan choti agricultural banks ko unke cyclical farming loans/deposits manage karne ke liye diya jata hai.
5. **Reserve Requirement:** Yeh fraction hai jo banks ko apne deposits ka rakhna zaroori hota hai.
6. **Operating Instruments:** Yeh tools hain jo central bank directly control karta hai, jaise interest rates ya monetary base.
7. **Intermediate Targets:** Yeh instruments hain jo central bank ke control mein nahi hote, magar uske tools aur final objectives ke

darmiyān hote hain.

Key Points

- 1. Federal Funds Rate Control:** Central bank reserves ki quantity ko open market operations se manipulate kar ke federal funds rate ko target tak le ata hai.
- 2. Three Types of Discount Loans:** Primary, secondary, aur seasonal credit, har ek ka apna purpose hota hai.
- 3. Reserve Requirement Limitation:** Yeh tool zyada useful nahi hai kyunke choti changes bhi deposits par bara asar dalti hain, isliye yeh aaj sirf reserve demand stabilize karne ke liye use hota hai.
- 4. Three Desirable Instrument Features:** Observable, controllable/quickly changeable, aur policy objectives se tightly linked hona.
- 5. Abandoned Intermediate Targets:** Central banks ne intermediate targets (jaise monetary aggregates growth) ko chor diya hai aur ab seedha apne objectives par focus karte hain.

Superstructure

- 1. Operating Instruments:** Interest rates, monetary base.
- 2. Intermediate Targets:** Growth in monetary aggregates.
- 3. Final Objectives:** Low inflation, high growth.

Example

Socho central bank ko lagta hai ke banks ke darmiyān udhaar lene dene ki rate (jaise dost dost mein paisay lend karte hain) bohot zyada ho gayi hai. Tou central bank market mein securities khareed kar reserves barha deta hai, jisse banks ke pass zyada paisa ho jata hai aur woh kam rate par ek dusre ko lend karte hain — bilkul waise jaise agar market mein kisi cheez ki supply barh jaye tou uska price gir jata hai.

Conceptual Questions

Q: What is the difference between primary credit and secondary credit, and why does this distinction matter for financial stability?

Answer: Primary credit sirf sound institutions ko short-term basis par milta hai jab unki daily reserve demand pori nahi ho rahi hoti, jabke secondary credit un institutions ko milta hai jo sound nahi hain ya unhe longer-term problems hain.

Jaise agar koi healthy bank ko sirf ek din ke liye reserves ki kami ho, tou woh primary credit le sakti hai, magar agar koi bank financial trouble mein hai, tou usay secondary credit lena parega jo zyada scrutiny ke sath ata hai.

Q: Why have central banks largely abandoned intermediate targets like monetary aggregate growth in favor of directly targeting their final objectives?

Answer: Central bankers ne realize kiya ke intermediate targets aur final objectives ke darmiyān ka link (jaise monetary aggregates aur inflation) itna reliable nahi tha, isliye ab woh seedha apne operating instruments (jaise interest rate) ko apne final objectives (jaise low inflation) se link karte hain.

Jaise pehle central banks money supply growth ko target karte the yeh soch kar ke iska inflation par seedha asar hoga, magar yeh relationship itni predictable nahi nikli, isliye ab woh seedha interest rates ke through inflation aur growth ko target karte hain.

LESSON QUIZ

1. What does the central bank manipulate to control the federal funds rate? **(The quantity of reserves through open market operations)**
2. Name the three types of discount loans the central bank makes. **(Primary credit, secondary credit, and seasonal credit)**
3. Who primarily uses seasonal credit? **(Small agricultural banks)**
4. Why is the reserve requirement not used as a direct tool of monetary policy today? **(Because small changes in it have too large**

an impact on deposit levels)

5. What are the three desirable features of a good policy instrument? (**Easily observable, controllable and quickly changed, and tightly linked to policy objectives**)

LESSON 37

Why Do We Care About Monetary Aggregates?

Deep Explanation

Yeh bilkul waisi baat hai jaise tumhare class mein agar ek hi rupiya bohot tezi se ek dost se dusre dost ke pass jaye (jaise koi cheez khareedne ke liye), tou woh rupiya bohot zyada transactions kar sakta hai bina naya paisa print huay. Money bhi economy mein yeh hi tarah ghoomti hai, aur yeh ghoomne ki speed hi velocity kehlati hai.

Key Terms

- 1. Velocity of Money:** Yeh batata hai ke har rupiya ek certain time mein kitni baar payments banane ke liye use hota hai.
- 2. Equation of Exchange:** Yeh formula hai jo money, velocity, price level, aur real GDP ko jorta hai ($MV = PY$).
- 3. Quantity Theory of Money:** Yeh theory kehti hai ke agar velocity aur real output stable hon, tou money supply double karne se price level bhi double ho jata hai.
- 4. Money Demand (M_d):** Yeh nominal GDP ko velocity se divide karne se nikalti hai.
- 5. Money Supply (M_s):** Yeh central bank aur banking system ke behavior se determine hoti hai.
- 6. Monetary Accommodation:** Yeh tab hota hai jab money growth itni high ho ke sustained inflation ko support kar sake.

Key Points

- 1. Inflation-Money Growth Link:** Har high-inflation country mein high money growth hoti hai, isliye sustained high inflation monetary accommodation ke bina mumkin nahi.
- 2. Equation of Exchange Derivation:** Money x Velocity = Nominal GDP, jo Price Level x Real GDP ke barabar hota hai.
- 3. Friedman's View:** Milton Friedman ke mutabiq inflation ek monetary phenomenon hai, agar velocity aur real output stable rahein.
- 4. Percentage Change Form:** Money Growth + Velocity Growth = Inflation + Output Growth.
- 5. Real Growth Explanation:** Countries mein money growth inflation se zyada ho sakti hai agar woh real growth experience kar rahe hon.

Example

Socho Ali ke pass Rs. 100 cash hai jo woh Bilal se calculator khareedne ke liye deta hai. Bilal yeh paisa Chohan se cricket match ke tickets khareedne mein laga deta hai, aur Chohan yeh paisa Dilawer se pencils khareedne mein use karta hai. Sirf Rs. 100 ne total Rs. 300 ki transactions create kar di, kyunke har rupiya teen baar use hua — yahi velocity of money ka concept hai.

Conceptual Questions

Q: Why does the quantity theory of money conclude that doubling the money supply doubles the price level?

Answer: Yeh assumption is liye hai ke velocity aur real output mein koi important change nahi hoti, isliye agar money double ho jaye aur velocity/output same rahein, tou equation $MV=PY$ ke mutabiq price level (P) ko bhi double hona parega taake equation balance rahe.

Jaise agar M aur V dono fixed hain, aur Y bhi fixed hai, tou M ko double karne se P ko bhi double hona zaroori hai taake $MV=PY$ ka rishta theek rahe.

Q: Why can a country experience higher money growth than its inflation rate without this being a contradiction?

Answer: Agar koi country real growth experience kar rahi hai (yani uska real output barh raha hai), tou extra money growth us extra real output ko support karne ke liye use hoti hai, na ke sirf inflation create karne ke liye.

Jaise equation of exchange ke mutabiq, $\text{Money Growth} + \text{Velocity Growth} = \text{Inflation} + \text{Output Growth}$, agar output growth positive ho tou money growth inflation se zyada ho sakti hai bina extra inflation create kiye.

LESSON QUIZ

1. What does the equation of exchange state? (**$MV = PY$, where M is money, V is velocity, P is price level, and Y is real GDP**)
2. What is the velocity of money? (**The number of times each unit of currency is used in making payments per unit of time**)
3. According to the quantity theory of money, what happens to prices if money supply doubles (assuming constant velocity and output)? (**Prices double**)
4. Who is credited with the statement that inflation is a monetary phenomenon? (**Milton Friedman**)
5. Is it possible to have high, sustained inflation without monetary accommodation? (**No**)

LESSON 38

The Facts About Velocity

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apni pocket money ko bank mein rakhne ke bajaye thora thora apne pass cash mein rakho taake roz kharch kar sako, magar agar bank zyada interest de tou tum apna paisa bank mein hi rakhna pasand karoge. Yeh decision ke kitna paisa hath mein rakha jaye, yahi "money demand" ka concept hai jo velocity ko affect karta hai.

Key Terms

1. **Transactions Demand for Money:** Yeh paisa hai jo log apne daily transactions ke liye apne pass rakhte hain.
2. **Precautionary Demand for Money:** Yeh paisa hai jo log unexpected expenses ke liye apne pass rakhte hain.
3. **Portfolio Demand for Money:** Yeh demand hai jo money ko ek investment option ke tor par dekhti hai, bonds aur stocks ke against.
4. **Nominal Income:** Yeh income hai jo bina inflation adjustment ke measure ki jati hai.
5. **Service Charge:** Yeh cost hai jo funds ko checking aur bond account ke darmiyan shift karne par lagti hai.

Key Points

1. **Friedman's Recommendation:** Friedman ne suggest kiya ke central banks ko money growth ko constant rate par rakhna chahiye, banks ki discretion ko limit kar ke.
2. **Velocity Stability:** Long run mein velocity stable hoti hai, magar short run mein bohot zyada fluctuate kar sakti hai.
3. **Money Growth Formula:** $\text{Money Growth} = \text{Inflation} + \text{Real Growth} - \text{Velocity Growth}$.
4. **Three Money Demand Factors:** Nominal income, holding cost (interest foregone), aur substitutes ki availability.
5. **Interest Rate Effect:** Jaise interest rate barhta hai, log apna checking balance kam rakhte hain, jo velocity ko barha deta hai.
6. **Technology Effect:** Financial innovation (jaise automatic transfers) paisay ko shift karne ki cost kam karti hai, jo velocity ko barhati

hai.

Example

Socho tumhari monthly pocket money Rs. 30,000 hai jo tum roz Rs. 1,000 kharch karte ho. Agar tum apna pura paisa checking account mein rakho tou average balance Rs. 15,000 hoga. Lekin agar tum half paisa bond fund mein daal do aur mid-month mein wapas le aao, tou tumhara average balance kam ho jayega (Rs. 7,500), jisse tumhe extra interest milega, magar shifting ki cost (Rs. 20) bhi lagegi. Agar interest income service charge se zyada ho, tou shifting karna fayda mand hoga.

Conceptual Questions

Q: Why does a higher nominal interest rate lead to higher velocity of money?

Answer: Jab interest rate zyada hota hai, tou log apna paisa checking account mein rakhne ke bajaye interest-bearing assets (jaise bonds) mein shift karte hain taake interest income mil sake, jiski waja se woh transactions ke liye kam paisa apne pass rakhte hain, aur har rupiya zyada baar use hota hai.

Jaise example mein dekho, agar bond fund 0.27% se zyada monthly interest deta hai, tou log apna paisa shift karna shuru kar dete hain taake interest kama sakein, jo unke average money holding ko kam kar deta hai.

Q: How does financial technology (like automatic transfer accounts) affect the velocity of money?

Answer: Technology paisay ko bond fund aur checking account ke darmiyan shift karne ki cost kam kar deti hai, isliye log kam paisa apne pass cash ya checking mein rakhte hain aur baqi interest-bearing assets mein rakhte hain, jiski waja se velocity barh jati hai.

Jaise example mein dekho, automatic transfer account ki waja se average money holding Rs. 1,500 se Rs. 1,000 tak gir gaya, jo dikhata hai ke technology ki madad se log kam paisa idle rakhte hain.

LESSON QUIZ

1. According to Friedman, what should central banks do with money growth? **(Set it at a constant rate)**
2. Is velocity stable in the long run or short run? **(It is stable in the long run, with significant fluctuations in the short run)**
3. Name the three factors that determine transactions demand for money. **(Nominal income, the cost of holding money, and the availability of substitutes)**
4. What happens to money holdings and velocity when the nominal interest rate rises? **(Money holdings decrease and velocity increases)**
5. What is the term for money held to guard against unexpected expenses? **(Precautionary demand for money)**

The Portfolio Demand for Money

Deep Explanation

Yeh bilkul waisi baat hai jaise tum apna paisa kahan invest karo yeh decide karte ho — kabhi cash apne pass rakhte ho kyunke asaani se use ho sakta hai, kabhi bank mein deposit karte ho kyunke interest milta hai. Yeh decision depend karta hai ke dusre options kitne risky ya liquid hain.

Key Terms

- 1. Portfolio Demand for Money:** Yeh demand hai jo money ko ek investment option ke tor par dekhti hai, dusre assets jaise bonds ke against.
- 2. Potential Output:** Yeh economy ki woh production capacity hai jab resources normal rate par use ho rahe hon.
- 3. Output Gap:** Yeh farq hai jo current output aur potential output ke darmiyan paida hota hai unexpected events ki waja se.
- 4. Aggregate Demand (Y^d):** Yeh formula MV/P se nikalti hai, jo money, velocity, aur price level se related hai.

Key Points

- 1. Interest Rate Expectation Effect:** Agar interest rates future mein barhne ki ummeed ho, tou log bonds chor kar money hold karna shuru kar dete hain.
- 2. Risk and Liquidity Effects:** Agar dusre assets risky ya less liquid hon, tou money demand barh jati hai.
- 3. Two Criteria for Money Growth Targeting:** Monetary base aur money quantity ke darmiyan stable link, aur money quantity aur inflation ke darmiyan predictable relationship.
- 4. Long-Run Inflation Formula:** Inflation, velocity changes ko ignore karte huay, money growth minus potential output growth ke barabar hoti hai.
- 5. Aggregate Demand-Price Level Relationship:** Price level barhne se money ki purchasing power kam ho jati hai, jo aggregate demand ko kam kar deti hai.

Superstructure

- 1. Transactions Demand Determinants:** National income, interest rates, alternative payment methods ki availability.
- 2. Portfolio Demand Determinants:** Wealth, relative return, expected future interest rates, relative risk, relative liquidity.

Example

Socho tumhare pass kuch paisa hai jo tum bank mein rakh sakte ho ya cash mein apne pass. Agar tumhe pata chale ke aage interest rates barhne wale hain, tou tum abhi cash mein rakhna pasand karoge taake baad mein behtar rate par invest kar sako. Yahi portfolio demand for money ka concept hai jahan future expectations tumhare decisions ko affect karti hain.

Conceptual Questions

Q: Why does an expectation that interest rates will rise in the future increase the current demand for money?

Answer: Agar log sochte hain ke interest rates barhne wale hain, tou abhi bonds khareedna nuqsan ka baais ho sakta hai (kyunke bond prices interest rate barhne se girti hain), isliye woh apna paisa cash/money mein rakhte hain taake baad mein behtar rate par invest kar sakein.

Jaise agar tumhe pata ho ke aghay mahine interest rates barhne wale hain, tou tum abhi bond khareedne ke bajaye apna paisa wait karne ke liye cash mein rakhoge.

Q: What two criteria must be met for money growth to be used effectively as a direct monetary policy target?

Answer: Pehla yeh ke monetary base aur money ki quantity ke darmiyan ek stable link ho, aur dusra yeh ke money ki quantity aur inflation ke darmiyan ek predictable relationship ho.

Jaise agar central bank reserves barhata hai aur isse money supply predictable tareeqe se barhti hai, aur woh money supply phir predictable tareeqe se inflation ko barhati hai, tou money growth ek reliable policy target ban sakta hai.

LESSON QUIZ

1. What happens to money demand when interest rates are expected to rise in the future? **(It increases, as people shift from bonds to money)**

2. What is potential output? (**What the economy is capable of producing when resources are used at normal rates**)
3. In the long run, what does inflation equal (ignoring velocity changes)? (**Money growth minus growth in potential output**)
4. Name the two criteria needed for money growth to be used as a direct monetary policy target. (**A stable link between monetary base and money quantity, and a predictable relationship between money quantity and inflation**)
5. What happens to aggregate demand when the price level rises? (**It falls, because purchasing power of money decreases**)

LESSON 40

Money Growth, Inflation, and Aggregate Demand

Deep Explanation

Yeh bilkul waisi baat hai jaise agar kisi cheez ki qeemat barh jaye (jaise petrol), tou log apni spending kam kar dete hain kyunke unka paisa pehle jaisa kaam nahi karta. Yahi waja hai ke jab inflation barhti hai, tou aggregate demand (total spending) kam ho jati hai.

Key Terms

- 1. Aggregate Demand Curve:** Yeh curve batata hai ke inflation barhne se output kaise kam hota hai, isliye yeh downward sloping hota hai.
- 2. Real Interest Rate:** Yeh asal interest rate hai jo economic decisions ko influence karta hai, nominal rate se zyada important hota hai.
- 3. Consumption (C):** Yeh aggregate demand ka woh hissa hai jo households ke kharch se mutaliq hota hai.
- 4. Investment (I):** Yeh aggregate demand ka woh hissa hai jo businesses ke spending se mutaliq hota hai aur real interest rate se zyada sensitive hota hai.
- 5. Net Exports (NX):** Yeh exports minus imports hota hai, jo currency appreciation se affect hota hai.
- 6. Long-Run Real Interest Rate:** Yeh woh rate hai jis par aggregate demand potential output ke barabar ho jata hai.
- 7. Monetary Policy Reaction Curve:** Yeh batata hai ke policymakers inflation aur output gap ke response mein interest rates kaise set karte hain.

Key Points

- 1. Inflation-Aggregate Demand Relationship:** Agar inflation money growth se zyada ho, tou real money balances aur aggregate demand dono gir jate hain.
- 2. Real Interest Rate Lever:** Central bank nominal interest rate ko control kar ke real interest rate ko influence karta hai, jo real economy par asar dalta hai.
- 3. Four Aggregate Demand Components:** Consumption, Investment, Government Purchases, aur Net Exports ($Y^{ad} = C + I + G + NX$).
- 4. Real Interest Rate Impact:** Real interest rate barhne se Consumption, Investment, aur Net Exports teen tarah se gir jate hain, jo Aggregate Demand ko kam karta hai.
- 5. Long-Run Equilibrium:** Long-run real interest rate woh level hai jahan aggregate demand potential output ke barabar ho jati hai.

Example

Socho central bank ne interest rates barha diye. Ab logon ke liye loan lena expensive ho gaya, isliye businesses naye investments karne se ruk gaye (Investment fall), log apni saving barha kar consumption kam kar diya (Consumption fall), aur foreign investors ko Pakistan ke assets attractive lagne lage jisse currency strong ho gayi aur exports expensive ho gaye (Net Exports fall). Yeh sab milkar overall economy ki total spending (Aggregate Demand) ko kam kar dete hain.

Conceptual Questions

Q: Why is investment considered the most important component of aggregate demand that is sensitive to changes in the real interest rate?

Answer: Investment decisions seedha borrowing cost se linked hote hain — agar koi project ka internal rate of return borrowing ke cost se kam ho jaye (jaise real interest rate barhne se), tou woh project profitable nahi rahega, isliye investment seedha real interest rate changes se affect hota hai.

Jaise agar koi company factory expand karna chahti hai aur loan ka interest rate barh jaye, tou agar expected return loan ke cost se kam ho, tou company yeh project cancel kar degi.

Q: How does a rise in the real interest rate affect net exports through currency appreciation?

Answer: Jab real interest rate barhta hai, tou foreign investors ko us mulk ke financial assets attractive lagte hain, isliye unki demand barhne se local currency strong (appreciate) ho jati hai, jiski waja se exports expensive aur imports cheap ho jate hain, jo net exports ko kam kar deta hai.

Jaise agar Pakistan ke interest rates barh jayen, tou foreign investors Pakistani Rupee mein invest karna chahenge, jisse Rupee strong ho jayega aur Pakistani exports duniya ke liye expensive ho jayenge.

LESSON QUIZ

1. Why does the aggregate demand curve slope downward? **(Because higher inflation causes real money balances to fall, reducing aggregate demand)**
2. What rate, not the nominal interest rate, actually influences households' and firms' economic decisions? **(The real interest rate)**
3. Name the four components of aggregate demand. **(Consumption, Investment, Government purchases, Net exports)**
4. What happens to investment when the real interest rate rises? **(It falls, because the cost of financing rises)**
5. What defines the long-run real interest rate? **(The rate at which aggregate demand equals potential output)**

LESSON 41

Deriving the Monetary Policy Reaction Curve

Deep Explanation

Yeh bilkul waisi baat hai jaise koi strict teacher har choti si galti par fauran punishment de (jaldi react kare), jabke koi loose teacher chote masail ko ignore kar de aur sirf bare masail par react kare. Central bank bhi inflation mein change par yeh decide karta hai ke real interest rate ko kitni jaldi aur kitna change karna hai, aur yeh unki "aggressiveness" ko zahir karta hai.

Key Terms

1. **Monetary Policy Reaction Curve:** Yeh current inflation aur central bank ke set kiye gaye real interest rate ke darmiyan relationship hai.
2. **Target Inflation:** Yeh woh inflation rate hai jo central bank achieve karna chahta hai.
3. **Long-Run Real Interest Rate (r^*):** Yeh woh rate hai jis par aggregate demand potential output ke barabar hoti hai.
4. **Movement Along the Curve:** Yeh current inflation mein change ka reaction hai.
5. **Shift in the Curve:** Yeh real interest rate ke level mein change hai har inflation level par, jo target inflation ya long-run real interest

rate change honay se hota hai.

Key Points

- 1. Same Direction Response:** Policymakers inflation barhne par real interest rate barhate hain, aur inflation girne par kam karte hain.
- 2. Curve's Location:** Jab current inflation target inflation ke barabar ho, tou real interest rate bhi long-run real interest rate ke barabar hota hai.
- 3. Slope Determinant:** Curve ki steepness policymakers ki aggressiveness par depend karti hai — zyada aggressive policymakers ka curve steep hota hai.
- 4. Inflation Target Shift Effect:** Agar inflation target barhe, tou curve right ki taraf shift karta hai, har inflation level par real interest rate kam set hota hai.
- 5. Long-Run Rate Shift Effect:** Agar long-run real interest rate barhe (jaise government purchases barhne se), tou curve left ki taraf shift karta hai.
- 6. Aggregate Demand Connection:** Jab inflation barhta hai, central bank real interest rate barhata hai, jo consumption, investment, aur net exports ko kam kar ke aggregate demand ko kam karta hai.

Example

Socho ek central bank ka inflation target 2% hai aur long-run real interest rate 2.5% hai. Agar inflation 2% se 3% tak barh jaye, tou central bank real interest rate ko 2.5% se 3% tak barha deta hai (yani 1% inflation increase par 0.5% rate increase) — yahi curve par movement hai. Agar koi central bank zyada aggressive ho, woh shayad 1% inflation increase par 1% rate barha de, jo unka steeper curve dikhata hai.

Conceptual Questions

Q: What is the key difference between a "movement along" the monetary policy reaction curve and a "shift" of the curve itself?

Answer: Movement along the curve current inflation mein change ka reaction hota hai (curve waisa hi rehta hai), jabke shift tab hota hai jab target inflation ya long-run real interest rate change ho, jiski waja se har inflation level par real interest rate ka level hi change ho jata hai.

Jaise agar inflation 2% se 3% ho jaye aur central bank apna rate 2.5% se 3% kare, yeh movement hai; magar agar central bank apna inflation target hi 2% se 4% kar de, tou poora curve right ki taraf shift ho jayega.

Q: Why does a higher inflation target shift the monetary policy reaction curve to the right, lowering the real interest rate at every level of current inflation?

Answer: Agar central bank zyada inflation ko acceptable maan le (target barha de), tou woh kisi bhi given inflation level par pehle se kam aggressive react karega, isliye woh har inflation level par kam real interest rate set karega bajaye purane lower target ke mutabiq.

Jaise agar target 2% se 4% ho jaye, tou 3% inflation ko ab bhi target ke qareeb samjha jayega, isliye is par central bank zyada rate nahi barhayega jaisa pehle barhata tha jab target 2% tha.

LESSON QUIZ

1. What does the monetary policy reaction curve represent? **(The relationship between current inflation and the real interest rate set by policymakers)**
2. What happens to the real interest rate when current inflation equals target inflation? **(It equals the long-run real interest rate)**
3. What determines the slope of the monetary policy reaction curve? **(Policymakers' aggressiveness in keeping current inflation close to target)**
4. What happens to the curve when the central bank increases its inflation target? **(It shifts to the right)**

LESSON 42

The Aggregate Demand Curve

Deep Explanation

Yeh bilkul waisi baat hai jaise agar koi teacher har choti baat par fauran strict action le, tou students ka behavior bhi fauran change ho jata hai (zyada sensitive). Lekin agar teacher kam react kare, tou students ka behavior itna jaldi change nahi hota. Yahi tarah agar central bank inflation par bohot aggressively react kare, tou output bhi inflation ke changes se bohot zyada sensitive ho jata hai.

Key Terms

- 1. Aggregate Demand Curve:** Yeh curve batata hai ke current output current inflation ke changes par kitna sensitive hai.
- 2. Flat Aggregate Demand Curve:** Yeh batata hai ke output inflation se bohot zyada sensitive hai.
- 3. Steep Aggregate Demand Curve:** Yeh batata hai ke output inflation se kam sensitive hai.
- 4. Real Money Balances:** Yeh purchasing power hai jo inflation barhne se kam ho jati hai.
- 5. Aggregate Demand Shift:** Yeh curve ka poori tarah move ho jana hai, monetary policy reaction curve ke shift ya aggregate demand ke components ke change honay se.

Key Points

- 1. Three Sensitivity Factors:** Inflation ka real money balances par asar, policymakers ki reaction ki strength, aur interest rate ke against aggregate demand ka response size.
- 2. Inverse Curve Relationship:** Agar monetary policy reaction curve steep ho, tou aggregate demand curve flat hota hai, aur vice versa.
- 3. Two Downward Slope Reasons:** Pehla, inflation real money balances ko kam karti hai; dusra, inflation policymakers ko real interest rate barhane par majboor karti hai.
- 4. Additional Inflation Effects:** Inflation wealth ko kam karti hai, poor logon ko zyada affect karti hai, saving ko barhati hai, aur net exports ko kam karti hai.
- 5. Curve Shifting Causes:** Monetary policy reaction curve ke shift, ya aggregate demand ke non-interest-sensitive components (jaise government purchases, taxes, consumer confidence) mein change, curve ko shift karte hain.
- 6. Monetary Policy and Recessions:** Monetary policy recessions create bhi kar sakti hai aur unhe avoid bhi kar sakti hai, agar woh dusre sources se aane wale aggregate demand shifts ko neutralize kare.

Example

Socho do mukhtalif central banks hain. Pehla bank bohot aggressive hai aur inflation mein choti si tabdeeli par bhi interest rate ko bohot zyada barha deta hai — iski waja se output (jaise factories ka production) bhi bohot kam ho jata hai, isliye unka aggregate demand curve flat hoga. Dusra bank kam react karta hai, isliye unka output kam affect hota hai, aur unka curve steep hoga.

Conceptual Questions

Q: Why does a steeper monetary policy reaction curve result in a flatter aggregate demand curve?

Answer: Agar policymakers inflation mein chote change par bhi real interest rate ko bohot zyada barhate hain (steep reaction

curve), tou consumption, investment, aur net exports bohot zyada gir jate hain, jiski waja se output inflation ke changes se bohot sensitive ho jata hai, jo flat aggregate demand curve dikhata hai.

Jaise agar central bank 1% inflation increase par 2% rate barhaye (aggressive), tou investment bohot zyada gir jayega bajaye us central bank ke jo sirf 0.5% rate barhata hai.

Q: Why can monetary policy potentially both cause and prevent recessions?

Answer: Agar monetary policy reaction curve shift ho (jaise central bank apna inflation target kam kar de ya long-run real interest rate barh jaye), tou aggregate demand curve bhi left shift ho sakta hai jo output ko kam kar ke recession create kar sakta hai; lekin agar policymakers samjhdari se in shifts ko neutralize karen, tou woh recession ko rok bhi sakte hain.

Jaise agar government spending kam ho jaye (jo aggregate demand ko left shift karega), tou central bank interest rates kam kar ke is asar ko neutralize kar sakta hai aur recession se bach sakta hai.

LESSON QUIZ

1. What three factors influence the sensitivity of current output to inflation? **(The effect of inflation on real money balances, the extent of policymakers' reaction, and the response of aggregate demand to interest rate changes)**
2. If the monetary policy reaction curve is steep, what shape will the aggregate demand curve be? **(Flat)**
3. Name two reasons why the aggregate demand curve slopes downward. **(Higher inflation reduces real money balances, and higher inflation induces policymakers to raise the real interest rate)**
4. What happens to the aggregate demand curve when government purchases increase? **(It shifts to the right)**
5. Can monetary policy cause recessions? **(Yes, through shifts in the monetary policy reaction curve)**

LESSON 43

The Aggregate Supply Curve

Deep Explanation

Yeh bilkul waisi baat hai jaise ek dukaandar apni saman ki prices fauran change nahi karta, balke pehle wait karta hai aur dekhta hai ke demand zyada hai ya kam, phir ahista ahista prices adjust karta hai. Economy mein bhi inflation fauran change nahi hoti, balke gradually adjust hoti hai — isliye short run mein yeh "sticky" rehti hai.

Key Terms

1. **Aggregate Supply Curve:** Yeh batata hai ke economy aggregate demand curve par kahan settle hogi, inflation aur output ke darmiyan relationship explain karte huay.
2. **Inflation Persistence:** Yeh tendency hai jisme inflation slowly change hoti hai — agar low hai tou low rehti hai, agar high hai tou high rehti hai.
3. **Short-Run Aggregate Supply Curve (SRAS):** Yeh horizontal hota hai current inflation level par, kyunke inflation short run mein fixed rehti hai.
4. **Long-Run Aggregate Supply Curve (LRAS):** Yeh vertical hota hai us point par jahan current output potential output ke barabar ho.
5. **Output Gap:** Yeh farq hai current output aur potential output ke darmiyan.
6. **Inflation Shock:** Yeh production cost mein change hai jo SRAS ko shift kar deta hai.
7. **Expected Inflation:** Yeh log jo inflation future mein expect karte hain, jo current inflation ko bhi affect karti hai.

Key Points

- 1. Two Persistence Reasons:** Log expected inflation ke mutabiq prices/wages adjust karte hain, aur sab decisions ek hi waqt par nahi hote (staggered adjustments).
- 2. Firms' Short-Run Behavior:** Short run mein firms prices change nahi karte, balke quantities adjust karte hain.
- 3. Output Gap Effect on SRAS:** Jab current output potential output se zyada ho, SRAS upward shift karta hai; jab kam ho, SRAS downward shift karta hai.
- 4. Inflation Shock Effect:** Positive inflation shock (jaise energy prices barhna) SRAS ko upward shift karta hai, jo inflation barhati hai.
- 5. Long-Run Equilibrium Conditions:** Current output = potential output, aur current inflation = expected inflation.
- 6. Potential Output Growth:** Investment aur technology ki waja se potential output barhta rehta hai, jo LRAS ko right shift karta hai.

Example

Socho ek baker apni roti ki price har roz nahi badalta, kyunke usay pata hai ke customers confuse ho jayenge aur shayad usay galat samjhein. Agar demand achanak barh jaye, woh pehle zyada roti banata hai (quantity adjust karta hai) bajaye price barhane ke, aur sirf agar demand lambay time tak high rahe tou ahista ahista price barhata hai. Yahi short-run aggregate supply curve ka concept hai jahan firms quantity adjust karti hain, price nahi.

Conceptual Questions

Q: Why does inflation persistence cause the short-run aggregate supply curve to be horizontal rather than upward sloping?

Answer: Inflation persistence ka matlab hai ke current inflation short run mein tabdeel nahi hoti (fixed rehti hai), isliye firms apni price increase rate ko adjust nahi karte balke sirf quantities produce ki rate adjust karte hain, jiski waja se SRAS horizontal hota hai current inflation level par.

Jaise agar current inflation 5% hai, tou short run mein yeh waisi hi rahegi, aur sirf output (quantity) change hogi demand ke mutabiq, na ke inflation rate.

Q: What is the key difference between how an output gap and an inflation shock each cause the SRAS curve to shift?

Answer: Output gap (current aur potential output ka farq) inflation ko gradually adjust karta hai jab firms zyada ya kam demand dekh kar apni prices change karte hain, jabke inflation shock production cost mein achanak change (jaise energy prices) ki waja se hota hai jo seedha SRAS ko shift kar deta hai.

Jaise agar economy potential output se zyada produce kar rahi ho, tou SRAS gradually upward shift hoga, magar agar achanak oil prices barh jayen, tou yeh ek inflation shock hoga jo seedha SRAS ko upward shift kar dega.

LESSON QUIZ

1. What is inflation persistence? **(The tendency for inflation to change slowly over time)**
2. Why is the short-run aggregate supply curve horizontal? **(Because inflation is fixed/persistent in the short run)**
3. What happens to the SRAS curve when current output rises above potential output? **(It shifts upward)**
4. What is an inflation shock? **(A change in the cost of producing output, often due to energy prices)**
5. At what point is the long-run aggregate supply curve vertical? **(Where current output equals potential output)**

Equilibrium and the Determination of Output and Inflation

Deep Explanation

Yeh bilkul waisi baat hai jaise agar tum apni body se zyada exercise kar lo (capacity se zyada), tou jism dard mehsoos karne lagta hai aur tumhe normal level par wapas anay ke liye rest karna parta hai. Economy bhi waisi hi hai — agar output potential se zyada ho jaye, tou inflation barhne lagti hai jab tak woh wapas normal level par na aa jaye.

Key Terms

- 1. Short-Run Equilibrium:** Yeh point hai jahan aggregate demand curve aur short-run aggregate supply curve intersect karte hain.
- 2. Expansionary Output Gap:** Yeh tab hota hai jab current output potential output se zyada ho.
- 3. Recessionary Gap:** Yeh tab hota hai jab current output potential output se kam ho.
- 4. Self-Correcting Mechanism:** Yeh process hai jisme economy khud ba khud long-run equilibrium ki taraf wapas chali jati hai.
- 5. Stagflation:** Yeh woh situation hai jisme inflation barh jati hai aur output gir jata hai, ek positive inflation shock ki waja se.

Key Points

- 1. Output Gap and Inflation:** Expansionary gap inflation ko upward pressure deta hai, jabke recessionary gap inflation ko downward pressure deta hai.
- 2. Aggregate Demand Shift Effect:** Agar aggregate demand right shift kare (jaise government purchases barhne se), output pehle barhta hai magar inflation gradually barhti hai jab tak SRAS adjust na ho jaye.
- 3. Long-Run Equilibrium Conditions:** Current output = potential output, aur current inflation = target inflation = expected inflation.
- 4. Permanent Inflation Change:** Agar inflation permanently barh ya gir jaye, tou yeh sirf monetary policy ka result ho sakta hai (inflation target change honay ki waja se).
- 5. Inflation Shock Temporary Effect:** Inflation shock long-run equilibrium ko change nahi karta, sirf temporary stagflation create karta hai jab tak economy wapas normal na ho jaye.

Example

Socho ek factory normal se zyada production karna shuru kar deti hai (jaise extra orders ki waja se). Shuru mein output barh jata hai magar prices waisi hi rehti hain. Magar jab yeh extra production lambay time tak chalti hai, tou workers aur suppliers apni prices barhana shuru kar dete hain (inflation barhti hai), jab tak factory wapas apni normal capacity (potential output) par na aa jaye.

Conceptual Questions

Q: Why does the economy have a "self-correcting mechanism" when current output deviates from potential output?

Answer: Agar output potential se zyada ho, tou inflation upward pressure ki waja se barhti hai, jo SRAS ko upward shift karti hai jab tak output wapas potential level par na aa jaye; agar output kam ho, tou yeh process reverse mein chalta hai. Yeh dikhata hai ke economy khud ba khud apne long-run equilibrium ki taraf wapas chali jati hai bina kisi external intervention ke.

Jaise agar economy boom kar rahi ho (output > potential), tou inflation gradually barhegi jab tak central bank ya khud market forces output ko wapas normal level par na le aayen.

Q: Why does a positive inflation shock (like an oil price increase) cause "stagflation" instead of simply raising inflation or simply lowering output?

Answer: Inflation shock SRAS ko upward shift karta hai, jiski waja se ek hi waqt mein inflation barh jati hai (production cost zyada honay ki waja se) aur output gir jata hai (kyunke firms kam produce karte hain zyada cost ki waja se), isliye dono cheezein ek sath hoti hain — yahi stagflation hai.

LESSON QUIZ

1. What determines short-run equilibrium? **(The intersection of the aggregate demand curve and the short-run aggregate supply curve)**
2. What happens to inflation when there is an expansionary output gap? **(It rises, putting upward pressure that shifts the SRAS curve upward)**
3. What can cause a permanent change in inflation? **(A change in monetary policy, specifically the central bank's inflation target)**
4. What is "stagflation"? **(A situation of higher inflation and lower output, caused by a positive inflation shock)**
5. Does an inflation shock change the economy's long-run equilibrium point? **(No, only a change in potential output or the inflation target can do that)**

LESSON 45

Shifts in Potential Output and Real Business Cycle Theory

Deep Explanation

Yeh bilkul waisi baat hai jaise agar ek student apni padhne ki capacity barha le (jaise nayi technique seekh le), tou shuru mein uska result fauran change nahi hota, magar waqt ke sath woh apni nayi capacity ke mutabiq behtar perform karta hai. Economy bhi waisi hi hai — potential output barhne ka asar fauran nahi hota, balke gradually nazar ata hai.

Key Terms

1. **Real Business Cycle Theory:** Yeh theory kehti hai ke business cycle fluctuations potential output ke shifts se hote hain, na ke aggregate demand se.
2. **Recessionary Output Gap:** Yeh tab hota hai jab potential output barh jaye aur current output usse kam reh jaye.
3. **Stabilization Policy:** Yeh policy hai jisme central bank aggregate demand ko shift kar ke output ya inflation ko stabilize karta hai.
4. **Fiscal Policy:** Yeh government ki taxes aur spending decisions hain jo stabilization ke liye use ho sakti hain.
5. **Productivity:** Yeh capital aur labor ki efficiency hai jo potential output ko barhati hai.

Key Points

1. **Potential Output Shift Effect:** Jab potential output barhta hai, shuru mein output/inflation same rehte hain, magar waqt ke sath recessionary gap create hoti hai jo inflation par downward pressure dalti hai.
2. **Real Business Cycle Theory Assumption:** Yeh theory maanti hai ke prices aur wages flexible hain, isliye SRAS irrelevant ho jata hai aur sirf aggregate demand shifts inflation change karti hain, output nahi.
3. **Policy Tradeoff:** Central bank inflation shocks ke baad ya output stabilize kar sakta hai ya inflation, dono ek sath nahi.
4. **Aggressive vs Cautious Response:** Aggressive response inflation ko jaldi target par le ati hai magar output mein zyada girawat ke sath, jabke cautious response output ko stable rakhti hai magar inflation mein deviation zyada hoti hai.
5. **Productivity Increase Opportunity:** Productivity barhne se policymakers bina recession create kiye inflation target kam kar sakte hain.
6. **Fiscal Policy Limitations:** Fiscal policy slow hoti hai aur politics se affected hoti hai, isliye stabilization ka kaam zyada tar central

Example

Socho ek factory mein nayi machinery aa jati hai jo production ko zyada efficient bana deti hai (productivity barh jati hai). Shuru mein workers ki output waisi hi rehti hai, magar dheere dheere factory zyada saman bana sakti hai (potential output barh jata hai) kam cost par, jo prices ko thora kam karne mein madad deta hai bina kisi recession ke. Yeh central bank ko mauka deta hai ke woh apna inflation target kam kar sake.

Conceptual Questions

Q: Why does real business cycle theory conclude that shifts in aggregate demand affect only inflation and not output?

Answer: Real business cycle theory yeh assume karti hai ke prices aur wages flexible hain, isliye SRAS itni jaldi adjust ho jati hai ke equilibrium hamesha potential output ke point par hi ho, isliye aggregate demand ka koi bhi shift sirf inflation ko change karega, output ko nahi, kyunke output already apne potential level par fixed hai.

Jaise agar yeh assumption sahi ho, tou agar government purchases barh jayen, tou sirf prices barhenge, factory apna production level nahi badlegi kyunke woh already apni full capacity (potential output) par kaam kar rahi hai.

Q: Why must central bankers choose between stabilizing output and stabilizing inflation when responding to an inflation shock?

Answer: Inflation shock ke baad, central bank monetary policy reaction curve ko shift kar sakta hai (aggregate demand ko shift kar ke) magar SRAS ko shift nahi kar sakta, isliye agar woh aggressively react kare tou inflation jaldi target par aa jayegi magar output bohot gir jayega, aur agar cautiously react kare tou output stable rahega magar inflation target se zyada lambay time tak deviate karegi.

Jaise agar oil prices barh jayen aur central bank fauran rates barha de, tou inflation jaldi control mein aa jayegi magar businesses ka production bohot kam ho jayega; agar woh ahista react kare, tou production stable rahega magar log lambay time tak high inflation face karenge.

LESSON QUIZ

1. What happens to the short-run aggregate supply curve immediately after an increase in potential output? **(Nothing — it remains unchanged in the short run)**
2. What assumption does real business cycle theory make about prices and wages? **(That they are flexible, so inflation adjusts rapidly)**
3. According to real business cycle theory, what does a shift in aggregate demand change — output or inflation? **(Inflation, not output)**
4. Can central bankers stabilize both output and inflation simultaneously after an inflation shock? **(No, they must choose one or trade off between them)**
5. Name one defect of fiscal policy as a stabilization tool. **(It works slowly, or it is difficult to implement effectively due to political factors)**

Mubarak Ho!

Aap ne MGT411 — Money and Banking ke tamam lessons mukammal kar liye hain. Yeh safar sirf notes parhne ka nahi tha, balke roz thodi himmat aur consistency jama karne ka tha. Ab waqt hai revision ka — jo seekha hai usay dohraiyе, apni tayari par yaqeen rakhiye, aur poora bharosa Allah par rakhiye.

Discipline wahi qaim rakhta hai jo apne maqsad ko Allah ke sath jorta hai. In sha Allah, yeh notes aapke exam ki taiyari mein roshni ka kaam karenge.

رَبِّ زِدْنِي عِلْمًا

Rabbi zidni 'ilma — “Ae Mere Rab, mera ilm barha de.”

اللَّهُمَّ لَا سَهْلَ إِلَّا مَا جَعَلْتَهُ سَهْلًا

Allahumma la sahla illa ma ja'altahu sahla — “Ae Allah, koi cheez aasan nahi siwaye us ke jisay Tu aasan kar de.”

Mehnat kabhi zaya nahi jati.

**Allah par bharosa rakhein, apni tayari par yaqeen rakhein,
aur In shaa Allah kamyabi aap ka muqaddar hogi.**